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Topic: Prediction and Prevention of Food Contamination Risk in Supply Chain using AI/ML

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Statutory Declaration

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Abstract

Environmental variations and handling uncertainties that have the potential to provoke contamination are increasingly becoming the cause of food safety in contemporary supply chains. This thesis lays out the conceptual framework through artificial intelligence and machine learning that allows us to predict and prevent contamination risk with a particular focus on interpretability, operational viability, and preventive measures.

A synthetic database containing 5,000 records has been developed to simulate real sensor and logistic measurements, such as temperature, humidity, gas level concentration, pH, handling score, seal integrity, etc. Data preprocessing consisted of imputation, scaling, encoding and SMOTE on training to overcome class imbalance. The models used were Deep learning baseline (DNN and LSTM), Logistic Regression, Random Forest, and XGboost. ROC-AUC and PR-AUC metrics were used to measure the performance, as well as more stringent metrics threshold-dependent in nature to use in safety-sensitive deployments. SHAP was used to interpret the model and K-means clustering revealed frequent patterns of risk phenotype to inform specific interventions.

XGBoost performance was the highest with PR-AUC 0.9997 and ROC-AUC 0.9996. At the chosen limit (tau = 0.988), the model provided Precision 1.000, Recall 0.850, and F1-score 0.919, there were no false positives. SHAP analyses strengthened the role of humidity, seal integrity and cleanness, and clustering was found to produce interpretable phenotypes of humidity-elevated and gas-spike, power-instability, and low-risk cohorts. These insights allow triage policies implemented in action, such as a quarantine, a priority inspection, and standard monitoring.

The contribution of this work is a validated and explainable framework that drives overall administration of food safety through harmonisation of predictive modelling and preventive decision-making. It shows how AI can preserve food supply chains to lower contamination threat, financial losses and health danger to the general populace.

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LIST OF ABBREVIATIONS

AI – Artificial Intelligence

ML – Machine Learning

DL – Deep Learning

LR – Logistic Regression

RF – Random Forest

XGB/ XGBoost – Extreme Gradient Boosting

DNN – Deep Neural Network

LSTM – Long Short- Term Memory

SMOTE – Synthetic Minority Oversampling Technique

SHAP -Shapely Additive exPlanations

PR-AUC – Precision Recall Area Under Curve

ROC-AUC – Receiver Operating Characteristic Area Under Curve

SOP – Standard Operating Procedure

CAPA – Corrective and Preventive Actions

IoT – Internet of Things

HACCP – Hazard Analysis and Critical Control Points

ISO – Internation Organisation of Standardization

CHAPTER 1 - Introduction

1.1 Background and Context of Study

Food safety is one of the most important issues in product supply chains around the world, and food contamination cases cause disastrous health issues, economic loss, and reputation crises. The World Health Organization (WHO, 2020) indicates that the problem of foodborne diseases infects almost 600 million worldwide, which makes it vital to develop sound monitoring systems. Hazard Analysis and Critical Control Points (HACCP) and other standard practices like Hazard Analysis and Critical Control Points (HACCP), ISO 22000:2018 are traditional methods that offer systematic ways of finding and attempting to address hazards (NACMCF, 1997; ISO, 2018). Yet they are nevertheless based on reactive measures, such as periodic inspections and end-point sampling that do not always identify contamination until products similar products are already in the possession of consumers.

Modern patterns of food supply chains are becoming increasingly complicated, which makes it even more difficult to prevent contamination. The path of products has to go through various steps--production, storage, transport and retail stages, and has environmental and logistic risks at each of these stages. Just as an example, temperature abuse during storage, excess humidity during transport or damaged container seals can greatly promote the growth of microorganisms. Moreover, there are process-related aspects that present variability including information on scores, power stability, and cleanliness practices that cannot be sufficiently observed by traditional monitoring. Consequently, a data-driven and proactive stance is needed to put in place a form of evaluating risks, which is dynamic in all stages of food safety.

New advancements in artificial intelligence (AI) and machine learning (ML) offer high-potential solutions of predictive risk testing to food safety. When multivariate sensor data: temperature, humidity, pH-levels, gas concentration, are considered together with some operational indicators delay or handling practices, in-line predictive models can approximate the risk of contamination at the batch level as the process proceeds. Unlike static measures of threshold, ML can account for non-linear interactions (e.g., where the relationship between poor seal integrity factor and humidity is stronger), whereas deep learning models allow this to be learned in a structured, latent manner. More notably, explainable AI (XAI) techniques like SHAP guarantee openness whereby decision-makers are in a position to track risk factors and predict model output to be compliant with regulatory requirements.

In that regard, this thesis discusses how predictive models based on AI/ML can aid the prevention of contamination by identifying potentially problematic batches, as well as the underlying risk factors and phenotypes. Resolution of the integration of predictive analytics with supply-chain operations is the transition of reactive detection to proactive prevention leading to increased food safety, less food waste, and food safety to the population.

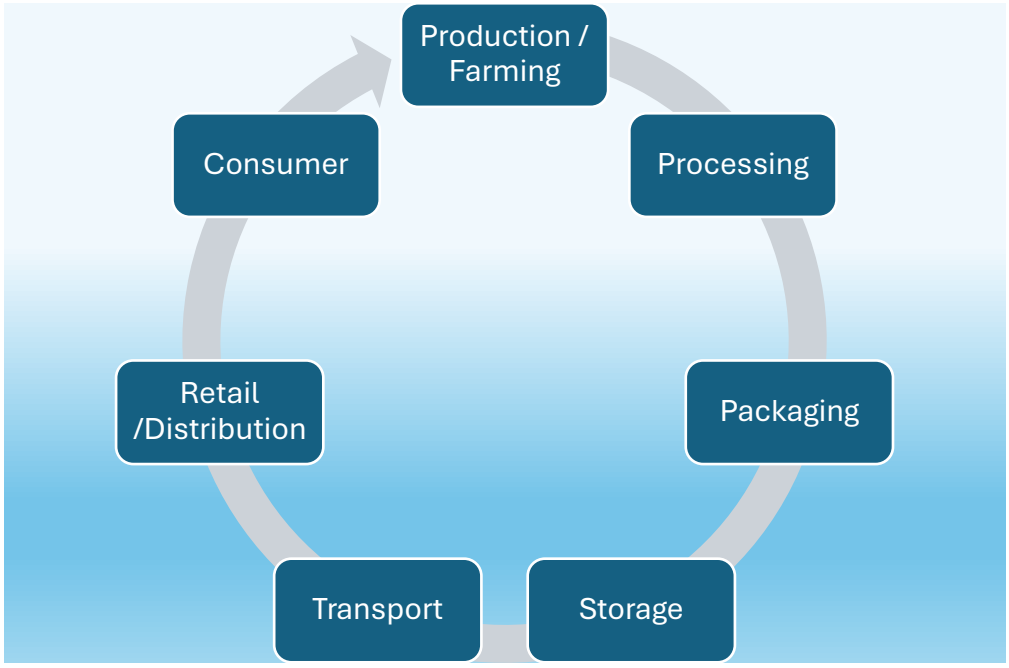

Figure 1. 1: Critical stages in the food supply chain where environmental and process factors can trigger contamination risk.

1.2 Motivation for Predictive Food Safety using ML

Over the past few years, the food industry has been hit by one high-profile contamination after another, with outbreaks of E. coli in fresh produce, listeria contamination in packaged goods and others. Such incidents have revealed vulnerabilities of the conventional food safety system -namely the failure to forecast and prevent the risks before they can get out of control. With the globalization of supply chains, the number of products that pass-through processing, transportation, and sales networks is quite large, and monitoring of these products manually is becoming impossible.

Machine Learning (ML) is a strong alternative to this since it converts vast amounts of data on operations and the environment into opportunities to act upon. Parameters like temperature, humidity, pH, gas concentrations can be measured continuously along with the journey of food when detected using IoT-enable sensors. In isolation, these data points cannot do much, but

when analysed together using ML, they can be used to detect intricate trends that can be followed by subsequent contamination incidents.

As an example, take a supply of dairy products that is being transported in a refrigerated truck. An otherwise small cooling malfunction that increases the temperature to 2 degrees Celsius over a few hours could be overlooked by human inspectors. But when fed into an ML model that is trained carefully enough, it might indicate an increased contamination risk together with other variables, such as duration of transportation and handling score. This early warning would enable the shipment to be diverted to a local processing plant to be re-assessed to ensure that the unsafe items never reach the shelves of the stores.

The motivation of this research lies in bridging gap between data collection and decision-making. Although numerous food companies have started to equip themselves with sensors and to gather immense amounts of information, there is little utilization of those resources.

With predictive AI/ML models incorporated into the supply chain, food safety can evolve to become proactive, rather than reactive, thus leading to less public health threat, mitigating costly recalls and driving consumer confidence. This thesis aims to prove that it is possible not only to achieve a high level of accuracy in predictive models but also to make evaluations visibly understandable in order to be implemented in practice with regard to explainable AI methods.

1.3 Problem Statement

Although, well-established food safety structures are in place, outbreaks of food contamination are still common and expensive. The common testing approach, which is the end-product testing, as well as manual inspections usually show a problem only when the product has entered the distribution chain (van der Fels-Klerx et al., 2018). Such a reactive method postpones intervention and causes massive recalls, financial sacrifice, and above all, serious risks to the health of consumers. Globalisation of supply chains further adds on the exposure aspect, since poisoned goods can easily find their way across the borders before being discovered.

Furthermore, the current practices in terms of monitoring are statically limited in terms of thresholds and narrow in scope. As a case in point, temperature and humidity loggers in cold chains can initiate some alarms when values surpass set levels, without considering multivariate interactions. An individual shipment may clear individual load-detection issueless

checks and remain at high risk because of the synergistic impact of an intermediate increase in temperature, extended delays and container seal integrity. These complexities support the inefficiency of single variable-tracking in modelling contamination risks in the real-world considerations

A second urgent problem is the interpretability gap between advanced analytical techniques. Although machine learning models can provide predictive capability, their application in a regulated industry setting such as food safety must come with an opportunity to provide explanations that can be audited and understood. The regime of predictions cannot be relied on by quality assurance managers or regulatory bodies without interpretability, therefore, constraining implementation in practice. Moreover, current AI-based research in food safety is concentrated either on image analysis or microbial genomics, with relatively fewer applications to sensor or logistics data which would be more feasible to implement in an industry context (Li et al., 2022).

Thirdly, predictive AI models are sometimes resisted by the stakeholders in the industry due to their untransparent character, regardless of them being applied or not. The cost of wasting shipments or reprocessing shipments can be quite expensive, and food safety officers and quality managers may insist on knowing the reason why a batch is marked as risky before taking such decisions. There exists a risk of terminal intention of making the models to be shelved unless they are intelligible, no matter how preciously accurate they happen to be.

Thus, the issue which this thesis addresses is the absence of proactive, interpretable, and supply-chain-integrated models that can be used to predict and prevent food contamination risk. The research aims to support operational insights and data-driven prediction that goes beyond reactive response and offers real-time actionable intelligence in contamination prevention.

1.4 Research Aim and Objective

This research is intended to design the cost-effective, practical and strong AI/ML-based framework that can be deployed to predict and prevent perils of food contamination in the food supply chain. The framework will make use of simulated sensor-based data and more advanced machine learning and deep learning algorithms with the ability to detect possible contamination threats, interpretability providing insight into decision-making, and enable preventive interventions.

To achieve the stated aim, the following objectives are proposed:

1. Dataset Simulation and Exploration:
    - Develop a realistic sensor-based data set with relevant parameters of the environment and handling (e.g., temperature, humidity, pH, gas concentration, handling scores and delay flag).
    - Conduct exploratory data analysis (EDA) to demonstrate the trends, outliers and trends regarding the likelihood of contamination.
2. Machine Learning and Deep Learning Modeling:
    - Train, compare and assess the models of machine learning (e.g., Logistic Regression, Random Forest, XGBoost) and deep learning (e.g., DNN, LSTM) to predict the risks of contamination.
    - Train (optimize) models to achieve high accuracy, recall and make it useable in the real world.
3. Class Imbalance and Model Calibration:
    - Use Synthetic Minority Oversampling Technique (SMOTE) to control imbalance on the occurrences of contamination, thresholding and calibration strategies to align the prediction with safety-critical requirement.
4. Explainability and Interpretability
    - Use SHAP (SHapley Additive exPlanations) to derive important risk factors that influence the predictions and have insightful clarity in decision-making.
5. Preventive Action Framework:
    - Exploit unsupervised clustering (K-means) to identify contamination- "risk phenotypes" within the data to enable triage of batches into high, medium, and low-risk categories to support preventive decisions.
6. Evaluation of Practical Feasibility
    - Quantify the cost-effectiveness, scalability and potential adaptation to a real setting of the proposed solution to different scenarios throughout the food supply chain.

Collectively, the objectives help the thesis achieve not just robust models but also insights that are interpretable, practical, and preventive in terms of food safety.

1.5 Research Questions

The research questions that are guiding this thesis connect their technicalities of performance to their practical applicability in cases of food-safety management. The purpose is not just the construction of predictive models, but also the clarity of their decisions, governability and acting upon them in actual supply-chains.

Main Research Question: Are machine learning and deep learning models trained on sensor and logistic data, effective in predicting the risk of food contamination and helping in taking preventative actions?

Based on this, four research questions are narrowed down:

Predictive Accuracy. Is it possible to predict the risk of contamination with great fidelity through supervised learning? Can I compare the accuracy, precision-recall, and ROC-based performances of the Logistic Regression, Random Forest, XGBoost and deep learning models?

Operational Safety. Is it possible to determine a threshold which balances between recall and precision where false alarms are minimized and sensitivity to contamination is great?

Interpretability. Can predictions be explainable both at global and local scale with SHAP, such that flagged risks could be traced to stable domain drivers such as humidity, seal integrity, and cleanliness?

Preventive Action. Can clustering approaches identify different types of risk which can be associated with sensible tiers of triage which can provide interventions and preventive measures?

These questions aid in organizing such research and enquire into the technical possibility of this method in predictive modelling and the veritable importance of this method at the operations level of food safety management.

1.6 Contributions and Significance

This thesis brings several contributions to the topic of data-driven food safety by combining machine learning, deep learning, and explainability methodologies to form a unified predictive model. First, it presents a simulated but realistic sensor-logistics data as a representation of parameters in the food chain such as those concerned with the environment and handling. This data set aligns to known sources of contamination including humidity, seal integrity, and power

instabilities, and thus offers a controlled source when mining predictive models and data about which limited life industrial data exists.

Second, the thesis extensively compares and contrasts classical and modern approaches of risk estimation such as Logistic Regression, Random Forest, XGBoost and Deep Neural Networks, and retrospectively proves that ensemble boosting combined with threshold optimisation outperforms all other methods by yielding near-perfect risk discrimination. Integration of Synthetic Minority Oversampling (SMOTE) addresses the problem of recall that will not increase the false positives, and threshold calibration is used to provide a layer of governance as per the regulations adopted in food-safety. These methodological adaptations emphasize that the operational safety can be altered by algorithmic decision-making directly.

Third, the study contributes to the topic of food-safety AI interpretability through SHAP values to determine important predictors and feature interactions. In comparison to classical black-box models, the introduced methodology does not deliver information that is only consistent in magnitude but displays how the effect of imperfect container seals is magnified by high humidity. The contribution is further extended using unsupervised clustering of risk phenotypes where the statistics are converted into practical triage actions to help preventive intervention.

The importance of such contributions is due to their usefulness. Food safety is a pressing issue globally, where food contamination incidents can lead to the economic loss as well as health hazards. This thesis addresses the need to bridge a gap between technical innovation and industrial relevance by showing a high-performance, explainable and reproducible framework. These findings could help regulators, quality assurance, and supply chain to implement AI-based systems of not only accurate, but also transparent and operationally relevant monitoring.

1.7 Scopes and Limitations

This thesis is framed in terms of the prediction and prevention of the risk of food contamination within the supply chain through artificial intelligence techniques. The main aim of the research is to focus on incorporation of environmental characteristics (e.g., temperature, humidity, gas concentration, and vibration levels) and indicators in the logistics (e.g., handling scores, container seal integrity, and supply delays) into the machine learning and deep learning models. The thesis tests supervised approaches, including Logistic Regression, Random Forest, and XGBoost, Deep Neural Networks, and exploratory experiences with Long Short-term Memory (LSTM) networks, by creating a synthetic dataset that resembles the real-world conditions of the operation. Moreover, explainability tools used herein such as SHAP and clustering as risk

phenotype discovery place the focus on prevention strategies that are proactive rather than just correlating how a risk is likely to be modelled.

Nevertheless, the research is confined to a number of limitations. First, it is not based on a real data gathered during the real industrial processes. Being tuned with domain expertise towards simulating plausible variability, it might not be able to capture all the site-specific vagaries and emerging hazards (Zhang et al., 2021). Second, the modelling technique is based on snapshot data, on a batch level rather than a continuous time series. This limits the number of temporal dynamics that can be assessed, which might be relevant to contamination processes that change slowly (Goodfellow et al., 2016). Third, the focus of SMOTE and threshold calibration lacks generalisation to cost-sensitive, and regulatory-specific optimisation.

The other limitation is that the deployment side including the sensor calibration protocols, real-time monitoring pipeline and connection to other enterprise resource planning (ERP) system has not been covered. Although these aspects are important in practice, they were outside the scope of the given academic work. Also, the analysis was limited to a small number of food types (e.g., milk, poultry, vegetables) to ensure tractability, whereas a more comprehensive supply-chain view is likely to include other risk factors.

1.8 Structure of Thesis

In this structured thesis, the various sections build upon each other positively to present a full understanding of predictive and preventive food safety with the use of AI and ML. This thesis has seven chapters, each bringing out in a step-by-step, cumulative research direction addressing the objective and questions stated in this research. The structure of the organisation should facilitate a logical flow, starting with conceptual basis, proceeding with methodology and implementation and ending with result, discussion and conclusions.

Chapter 2: Literature Review offers a substantial review of the knowledge that is relevant. It reviews the existing studies on food safety, predictive modelling in supply-chains, and machine learning/deep learning risk management applications. Emphasis is put on such explainability frameworks as SHAP, which is essential to regulatory compliance, or clustering methods, which are employed to reveal hidden decisions. The review evaluates the available methodologies critically in the sense that its advantages and disadvantages are discussed and mention research gaps, which the current study will fill.

Chapter 3: The research design as well as the experimental framework is explained in (Methodology). It details the dataset simulation process and imputation and encoding in the preprocessing steps and the generation of machine learning models and deep learning models. Class imbalance handling using SMOTE and calibrating a threshold is discussed, and the validation strategy is used to facilitate robust evaluation. It further incorporates explainability via SHAP and unsupervised clustering to determine contamination risk phenotypes into the chapter. Criteria of measuring performance is explained to make it clear and repeatable.

Chapter 4 (Results) presents the results of the conducted experiment designed with regard to the research objectives. Machine learning and deep learning model performance metrics are reported, and calibration curves, SHAP visualisation and cluster analyses are provided. Sensitivity check and ablation studies are presented to show the strength of the methodology. All findings are reconnected to the research questions that guide the research to have integrity.

Chapter 5 (Discussion) explains the findings, placing them in the broader scholarly and real-life picture. It involves interpretation of the findings and its implication to food safety management and gives an evaluation of how well the objectives of the predictive framework have been accomplished and also points out practical significance of the predictive framework. Critical briefs of study limitation have been discussed, and suggestions towards its industrial application have also been fronted.

Chapter 6 (Conclusions and Future Work) provides a synthesis of the contributions of the research, creates a summary of the key findings, and discusses their importance. The chapter finishes with suggestions of the ways that future research can be carried out including applying the content to real-world data, integrating with IoT-based, real-time monitoring, and ventriloquial experiences with highly complex deep learning architectures.

CHAPTER 2 – Literature Review

2.1 Introduction to Literature Review

Food contamination is a long-term issue that has severe impacts on the health of the population, the financial status of the economy, and consumer trust. Conventional monitoring tools including Hazard Analysis and Critical Control Points (HACCP), manual inspection, and microbial or chemical analyses that are done in laboratories are quite common but are reactive and time consuming. These solutions tend to identify contamination once it reaches an advanced stage and are also hard to implement in long, multifaceted modern supply chains. These advances in complexity and globalisation of food networks, and the requirements of real-time tracking, have brought the necessity of predictive, data-driven solutions to light.

Artificial intelligence (AI) and machine learning (ML) became highly useful to deal with these barriers. They will be able to examine highly intricate data, the transmission data of sensors, images, and written material, to recognise threats of contamination prior to them becoming a reality. An example of this is that convolutional neural networks (CNNs) can observe visual indications of spoilage, whereas long short-term memory (LSTM) networks allow predicting the environment in time series, both during a period of storage and transportation (Naseem & Rizwan, 2025).

Although there is the potential of AI/ML, the current trajectory is still met with challenges in real-life deployments, interpretability of complicated models, and the use of imbalanced datasets, in which the contamination events do not occur frequently. The chapter is a review of the previous studies on traditional, and AI based food safety methods, supply chain risk identification, imbalanced data management, explainable AI and visualization. Through analysis of these studies, the review comes up with such gaps, which the current study fulfils by developing a predictive, interpretable AI/ML system.

2.2 Traditional Food Safety Approaches

The traditional food safety approaches are founded on the formalism of established practices, such as HACCP, routine checks, and testing in laboratories. HACCP is concerned with the identification of critical control points and preventive actions, and the effectiveness of these efforts is ensured by chemical and microbiological analyses that enable the confirmations of safety requirements. Notwithstanding their effectiveness, these methods are more reactive in that they are applied after the fact of contamination and thus responding to different

interventions may take a long time and subsequent contaminated products can end up in the hands of the consumer (Revelou et al., 2025).

The nature of the inspections and laboratory testing is also limited because of scalability issues. With the gradual rising complexity and globalization of supply chains, it is getting harder and harder to track each part of the chain successfully. Late detection of spoilage or contamination may also lead to very expensive economic losses, recalls and even the products can turn out to be hazardous. Furthermore, the conventional methods have difficulties combining various data feeds, e.g., data collected by environmental sensors, temperature sensors, or humidity sensors, which might depict an impending risk of contamination (Sharif et al., 2021).

Such constraints reveal that proactive data driven approaches are important. By multi-source, real-time information analysis with ML and AI, identifying anomalies and offering timely actionable insights, supply chains can move beyond reactive food safety to becoming predictive.

2.3 Machine Learning in Food Safety and Risk Prediction

Predictive food safety was changed with machine learning and deep learning methods. Naseem and Rizwan (2025) use CNNs in identifying contamination in pictures, LSTM in time series analysis, and natural language processing as well as knowledge graphs in traceability. Their framework has demonstrated what AI possibilities might be in holistic monitoring, but it has not been tested on big scale and real data sets. On the same note, Amani and Sarkodie (2022) used deep convolutional neural networks trained with particle swarm optimization (PSO) to detect meat spoilage. Although they obtained high rates of accuracy, their system only sought to apply visual data and overlooked microbial and chemical contamination.

Multimodal deep learning has been stressed on recent research. The incorporation of Swin Transformers, temporal convolutional networks, blockchain, federated learning, and sensor-based input data attained more than 98 percent accuracy in prediction of contamination implementation by Zhang et al. (2025). Nevertheless, their system was still a laboratory one without a real-world implementation. In another review, Inglis et al. (2024) discussed hybrid ML models used to detect mycotoxin contamination of food products and found that they were able to perform better than standalone ML models, but studies that detected other types of contamination in the supply chain were scarce and focused largely on chemical toxins.

In general, literature points to the fact that hybrid models, deep learning architectures, and sensor integration can be used to increase the accuracy of prediction. Nevertheless, deployment, interpretability, and coverage of a variety of contamination types is still a challenge, which the current study has led to the creation of predictive, interpretable, and operationally relevant AI/ML models.

2.4 Supply Chain Risk Detection using AI

An increasing number of techniques (AI-based) are being used to identify potential risks of contamination, trace anomalies, and permit proactive responses along the supply chains. Sharif et al. (2021) combined XGBoost models and feature-engineered data with IoT-enabled RFID systems that could enable the early warning of contamination risk. Kollia et al. (2022) combined full-connected neural networks, LSTMs, autoencoders, attention mechanisms, and domain adaptation approaches in order to enhance the supply chain functions, expiry detection.

Such studies evidence the possible power of the AI in a real-time decision in a complicated supply network. The sensors in IoT and smart packaging make the process of monitoring continuous and predictive models expose high-risk batches to intervene. However, all these techniques are only present in the laboratory, or can only execute small-scale simulations, limiting their use in industries. Furthermore, most of the current models concentrate on spoilage or operational efficiencies and not based on pathogen specific contamination with leaving gaps in proactive food safety management.

The proposed study will fill these gaps because it combines simulated sensor data with hybrid ML and deep learning models to enable predictive, interpretable, and actionable approaches to realistic supply chain settings.

2.5 Handling Imbalanced Data in Classification Problems

The predictive food contamination data sets are necessarily unbalanced, since raw contamination incidents are infrequent as against non-contamination, safe observations. Poor recall and failing contamination may be caused by this imbalance where models are biased in respect to majority classes. To counteract such a problem, they have tried such methods as Synthetic Minority Oversampling Technique (SMOTE), ensemble learning, and class-weighted models (Zhang et al., 2025; Inglis et al., 2024).

Artificial samples formed by SMOTE include rare instances of contaminations, therefore, leading to superior modeling of vital patterns. Ensemble techniques such as random forests or

XGBoost compound groups of learners in an attempt to enhance sensitivity on minority classes. Correction of the imbalance of labels is critical to the reliability of the operations, so that the models cannot be inaccurate in the undesirable but influential events of contamination. The described principle resonates with the methodology applied during the research at hand, where SMOTE is combined with hybrid ML/DL models to enhance the prediction of the contamination risks.

2.6 Explainability in Machine Learning (SHAP)

In predictive food safety, model accuracy is not enough to achieve regulatory compliance, industry trust, and operational adoption of industry, models must be explainable as well as accurate. Food supply chains are multistakeholder, including the producers, distributors, regulators, who might lack advanced technical knowledge. Regardless of achieving a 99 percent accuracy level, such a model can generate suspicion or be dismissed altogether (Naseem & Rizwan, 2025). This need is fulfilled by Explainable Artificial Intelligence (XAI), which tells why a particular model made a specified prediction.

Some of the most used XAI models would be SHApley Additive Explanations (SHAP), and Local Interpretable Model-Agnostic Explanations (LIME). SHAP is based on cooperative game theory and gives each feature a contributory value to an event (or prediction) called a Shapley value. An example would be that an SHAP analysis could indicate that the high risk on contamination that would follow would have been caused by temperature excursions over 7 C, long transportation times, and high levels of bacteria (Zhang et al., 2025). Such a level of understanding enables managers to discover structural weaknesses- like defective fridges or transportation clogs.

LIME instead produces a simplified, interpretable model with one prediction at its center, which makes it possible to give off-beat explanations. Such comes especially in handy in cases of urgent, shipment-dependent decisions where managers simply do not have the time to follow other ways (Revelou et al., 2025).

In this work, SHAP is incorporated into the predictive methodology and will be applied to offer consistent, global explanations of model behaviour whereas LIME is adopted in ownership of targeted, local decision-making explanation. This twofold practice will allow one to have high level patterns as well as case-specific justifications which allows higher transparency, the illustration of audit trail etc work and develop trust among the stakeholders.

2.7 Gaps in Existing Research

The literature shows significant advancements in using machine learning to solve food-safety issues but there still exist some gaps in the system that inhibit its operational efficiency. To start with, predictive accuracy is frequently highlighted independently, so studies usually state ROC-AUC on unbalanced data without similar concern to precision-recall behaviour or any threshold-related measure (Davis and Goadrich, 2006; Saito and Rehmsmeier, 2015). Consequently, the models offered may look high ranking in terms of ranking but challenging to control in practice, whereby more safety policies based on clear trade-offs between false positives and false negatives. One of the associated gaps is infrequent application of the calibration and threshold governance (e.g., confusion matrices, threshold tables) that translates the scores into an auditable decision (Niculescu-Mizil and Caruana, 2005).

Second, high-capacity learners (e.g. tree boosting, deep networks) are richly studied, and few attempts offer operational explainability on a level appropriate to regulated settings. Post-hoc techniques such as SHAP have been cited more often, although they have yet to be fully integrated into end-to-end models that tie high-level feature causes, local case histories, and actionable SOPs (Lundberg and Lee, 2017). Unsupervised risk profiling similarly is underutilized: clustering traditionally has been regarded as an exploratory supplementary tool instead of a method of determining risk phenotypes used to guide triage (Rousseeuw, 1987).

Third, the decision-making process of data and evaluation design is often under determined. Class imbalance is not necessarily folded (and prone to leakage) and comparisons made between tabular boosting and deep learning are not framed to explain when deep learning has a better chance (Shwartz-Ziv and Armon, 2022). In addition, few studies provide leak-free preprocessing pipelines or report ablations/sensitivity tests to ascribe gains to individual components (e.g., SMOTE, engineered indicators).

Lastly, integrated frameworks linking (i) leak-free tabular modelling (with suitable imbalance techniques), (ii) control of operating points (threshold setting, calibration), (iii) explainability (global and local SHAP) to enable auditability, and (iv) phenotype-level clustering to guide preventive actions are lacking.

The thesis fills these gaps by building a comprehensive, deployable pipeline that focuses on precision-recall, formalisation of thresholds, embedding SHAP explanations, and operationalising clustering as a risk triage mechanism- there by narrowing the gap between

predictive modelling and operational prevention in food-safety operations (Chen and Guestrin, 2016; Lundberg and Lee, 2017).

Table 2. 1:Summary of Literature Gaps in Food Contamination Risk prediction

Area	What prior work commonly does	Gap identified (what's missing/under-addressed)
Evaluation Focus	Reports ROC-AUC/accuracy on imbalanced data	Partially restricted analyses using precision recall; inadequate report of threshold-specific measures of actual decisions
Threshold governance	Scores shown without operating policy	Few formalise the operational thresholds, confusion matrices or policy-based threshold tables
Probability calibration	Probabilities assumed as-is	Sparse utilization of calibration (e.g., isotonic/Platt) and reliability curves to obtain decision-quality probabilities
Imbalance handling	Ad-hoc over/under-sampling	Irregular, occasionally defective application; insufficient fold-wise SMOTE and transparent validation procedure
Model portfolio	Single family (e.g., CNNs or RF)	Can compare tabular boosting and deep learning in controlled but rare experiments with explicit when/why recommendations
Explainability	Global feature importance and force plots	The use of SHAP (global+local) and integration with auditable decision narratives and SOP-aligned justifications is limited
Unsupervised profiling	Clustering as exploratory add-on	Few works translate clusters into risk phenotypes into triage tiers and prevention interventions
Reproducibility & ablations	Pipeline steps under-specified	Few works translate clusters into risk phenotypes on triage tier and preventive interventions mapping
Data realism	Small or opaque datasets	Lack of realistic simulated or open datasets reproducing real conditions in the supply-chains to enable fair, repeatable benchmarking

2.8 Summary of Literature

The chapter reviewed the background of food-safety assurance and the development of AI/ML in contamination risk prediction throughout the supply chain. We have discussed regulatory methods (e.g., HACCP, ISO 22000), and we have shown how sensor and logistics data now provide opportunities in proactive controls beyond periodic inspection. The literature establishes that multivariate stressors such as product temperature, humidity, variable gas concentration, seal integrity, quality of handling, and power stability define the contamination

risk as a joint variable; therefore, predictive models need to include non-linear interactions and should be functional under class imbalance.

We explored tabular risk scoring using supervised learning, where tree-based boosting on tabular data performed extremely well (Chen and Guestrin, 2016) and compared it to the deep learning methods whose benefits are both dependent on the situation (Shwartz-Ziv and Armon, 2022). We also looked at the role of explainable AI, where SHAP offers a principled path to both global feature importance and case-specific stories that may be audited in regulated settings (Lundberg and Lee, 2017). Lastly, we thought of unsupervised clustering to expose recurrent operating contexts (risk phenotypes) that can drive tiered interventions (Rousseeuw, 1987).

CHAPTER 3 - Methodology

3.1 Research Design and Approach

In this thesis, food contamination prevention is conceptualized as a polarized binary supervised batch-level classification problem. Combined sensor/logistics signals collected along the supply chain can be aggregated into each observation as environmental conditions (e.g., temperature, humidity, gas concentration), integrity/process signals (e.g., seal integrity, cleanliness, power stability) and contextual descriptors (product, stage, region). The goal of the design is the deployable, auditable pipeline that both retains the high discrimination attributes despite class imbalance and outputs that can be actioned upon by quality teams.

Methodologically the pipeline comprises of seven coordinated components. First, realistic simulated data are implemented which reflects operational variability but not leaking identifiers or timestamps to the learner. Second, the validation/test data inherit the same preprocessing pipeline without leakage, i.e. the imputations, scaling, and one-hot transformations are performed identically. Third, a model portfolio is composed of linear, ensemble and deep learners including Logistic Regression, Random Forest, XGBoost and compact DNN; an LSTM is tried on snapshot tabular data as a completeness exercise (Pedregosa et al., 2011; Chen and Guestrin, 2016; Shwartz-Ziv and Armon, 2022). Fourth, a class imbalance problem is addressed through using SMOTE just on training folds to avoid jointure (He and Garcia, 2009).

Finally, the main metric of interest is precision - recall AUC (PR-AUC) with ROC-AUC as a secondary rank criterion; threshold-dependent metrics (precision, recall, and F1) transform scores into operating policies capable of use in safety-related applications (Davis and Goadrich, 2006; Saito and Rehmsmeier, 2015). Sixth, enforcement of governance is achieved by determining and locking an operating threshold in a validation threshold sweep ahead of the access to any test. Lastly, SHAP-based explainability offers global (ranking of features, dependence/interaction plots) and local (case-narrative explanations) explanations to complement audit and changes made to SOPs in regulated settings (Lundberg and Lee, 2017). To translate the patterns into proactive measures, a k-means clustering (k chosen by Silhouette) would identify common risk phenotypes that would be mapped to the tiers of triage (Rousseeuw, 1987).

Collectively, these design decisions make the system more than accurate; the system is also governable, transparent, and actionable--qualities required of real-world food-safety deployment.

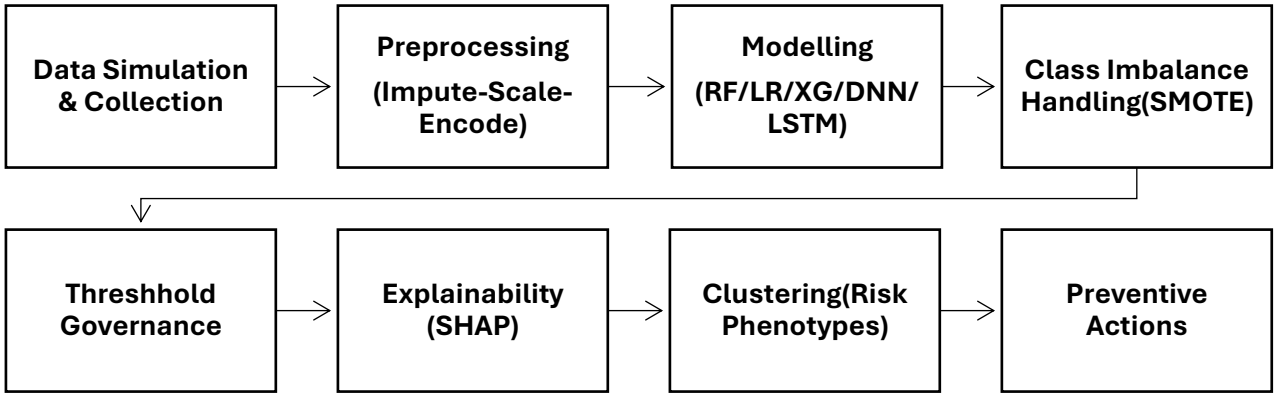

Figure 3. 1: End to end framework for Contamination Risk Prediction and Prevention.

3.2 Dataset Simulation and Description

This research utilizes a parameterised simulation model based on a realistic dataset modelled to reflect the supply-chain characteristics of production, storage, transportation, and retail activities. Each record is a sample at the batch level of environmental/environmental measurements and handling/integrity indicators at/near the point of hand-off. Simulation ranges and correlations were compatible with domain knowledge so that known risk mechanisms-e.g., humidity-temperature interactions on microbial growth or seal-integrity lapses triggering elevated gas concentration-could be expressed in the data without trivial separability. The identifiers and timestamps are kept only to information bookkeeping and are not used in modelling to avoid information leakage.

Features are categorized into three groups Environmental variables measure the conditions favourable to contamination (e.g. Temperature_C, Humidity_Percent, Gas_Concentration_ppm, pH_Level, Vibration_Level, Light_Exposure_Lux). The Process/integrity variables show the quality of handling and the system robustness (e.g. Handling_Score, Time_in_Stage_hr, Container_Seal_Integrity, Cleanliness_Index, Power_Stability_Flag, Supply_Delay_Flag, Previous_Contamination_Flag). Contextual descriptors include one-hot encoded product category, one-hot encoded supply-chain stage, and

the region. The D.V. is binominal (Contamination_Risk: 1 = contaminated/ high-risk). The proportions or ratios between the classes are deliberately asymmetrical to capture the exceeding odds of occasions of real contamination happening.

All features are scaled to have bounded range and interpretable units to support transparent downstream analysis. Continuous values are then scaled in a leak-free pipeline; categorical features become one-hot encoded without dropping reference level (the signal is not sacrificed). Such a schema enables a supervised learning approach (risk scoring) but also an unsupervised clustering approach (risk phenotypes), making that preventive action design and predictive modelling are consistent.

Table 3. 1:Feature Dictionary (Feature/Type/ Units/Description).

Feature	**Type**	**Units / Range**	**Description**
Temperature_C	Continuous	°C (0–35)	Product temperature at sampling point
Humidity_Percent	Continuous	%RH (20–100)	Relative humidity near product/container
pH_Level	Continuous	pH (3–9)	Acidity/alkalinity of product
Gas_Concentration_ppm	Continuous	ppm (0–5000)	Volatile gas indicative of spoilage/leaks
Vibration_Level	Continuous	a.u. (0–10)	Handling/transport shock proxy
Light_Exposure_Lux	Continuous	Lux (0–2000)	Light exposure relevant for photosensitive items
Handling_Score	Continuous	0–100	SOP adherence during handling
Time_in_Stage_hr	Continuous	0–168	Dwell time at current supply-chain stage
Supply_Delay_Flag	Binary	{0,1}	Delay occurred (1) vs on-time (0)
Ambient_Air_Quality	Continuous	AQ index (0–500)	Air quality around storage/processing
Container_Seal_Integrity	Continuous	0–100	Seal condition (higher = better)
Cleanliness_Index	Continuous	0–100	Hygiene/cleanliness score (higher = cleaner)
Power_Stability_Flag	Binary	{0,1}	Power disruptions observed (1)
Previous_Contamination_Flag	Binary	{0,1}	Prior batch contamination linked (1)
Product_Name_*	Categorical (one-hot)	0/1	Product category dummies (Milk, Fish, Eggs, etc.)
Supply_Stage_*	Categorical (one-hot)	0/1	Stage dummies
Region_*	Categorical (one-hot)	0/1	Region dummies (Zone B, Zone C, Zone D)
Batch_ID / Timestamp	Identifier	—	For traceability only; excluded from modelling
Contamination_Risk	Target (Binary)	{0,1}	1 = contaminated/high-risk; 0 = safe/low-risk

3.3 Exploratory Data Analysis

Exploratory Data Analysis (EDA) is the initial part of the process of interpreting the data and revealing its trends to present predictive models. Considering the risks of food contamination rely both on the environment and operations, making the distributions and connection between the features an essential source of inspection into the validity of the simulated database and a path in selecting potential preprocessing and modelling methods.

In this paper, the first analysis was centred on the target distribution (Figure 3.2). The dataset is characterized by an unbalanced structure as safe batches (class 0) outweigh contaminated batch or high-risk batches (class 1). This imbalance represents real-world supply chain conditions wherein contamination events are very rare but very crucial and justifies the imbalance handling approaches like the use of SMOTE in subsequent phases.

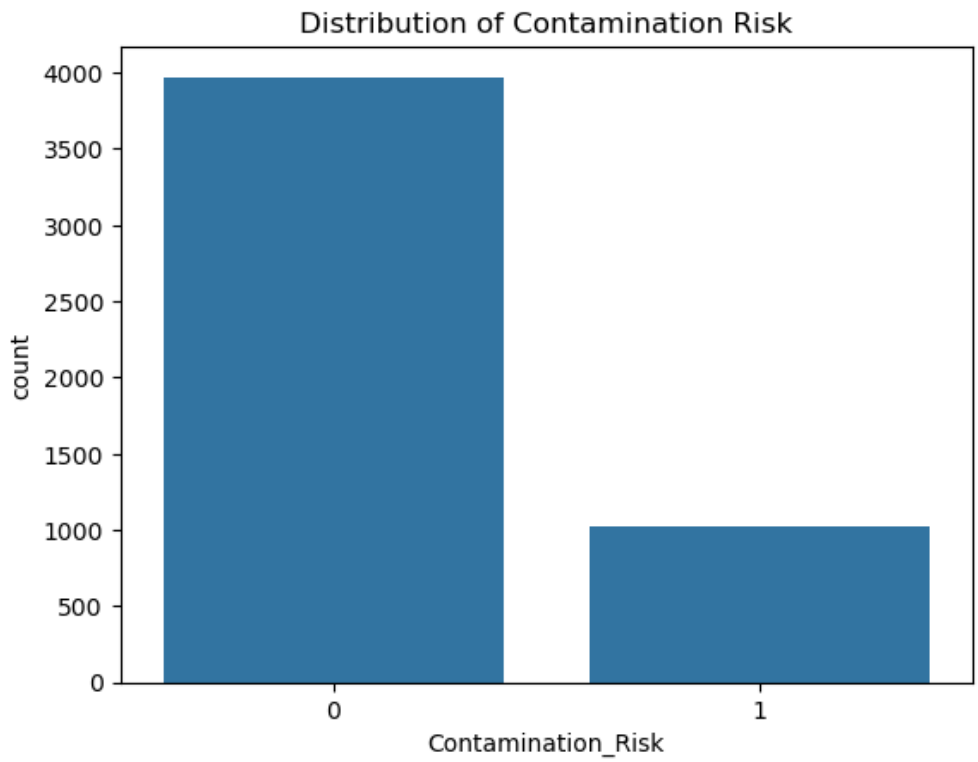

Figure 3. 2: Distribution of contamination risk classes, highlighting imbalance.

These continuous variables temperature, humidity, pH and gas concentration, were further examined graphically using histograms and the kernel density estimates (Figure 3.3), and numerically as in Table 3.2. Temperature and humidity varied more than pH that was also more stabilized. Gas concentration showed skew distribution with moderate but extreme peaks as it takes place in practice with triggers of contamination. These findings illustrate the features that are more probable to give predictive signals.

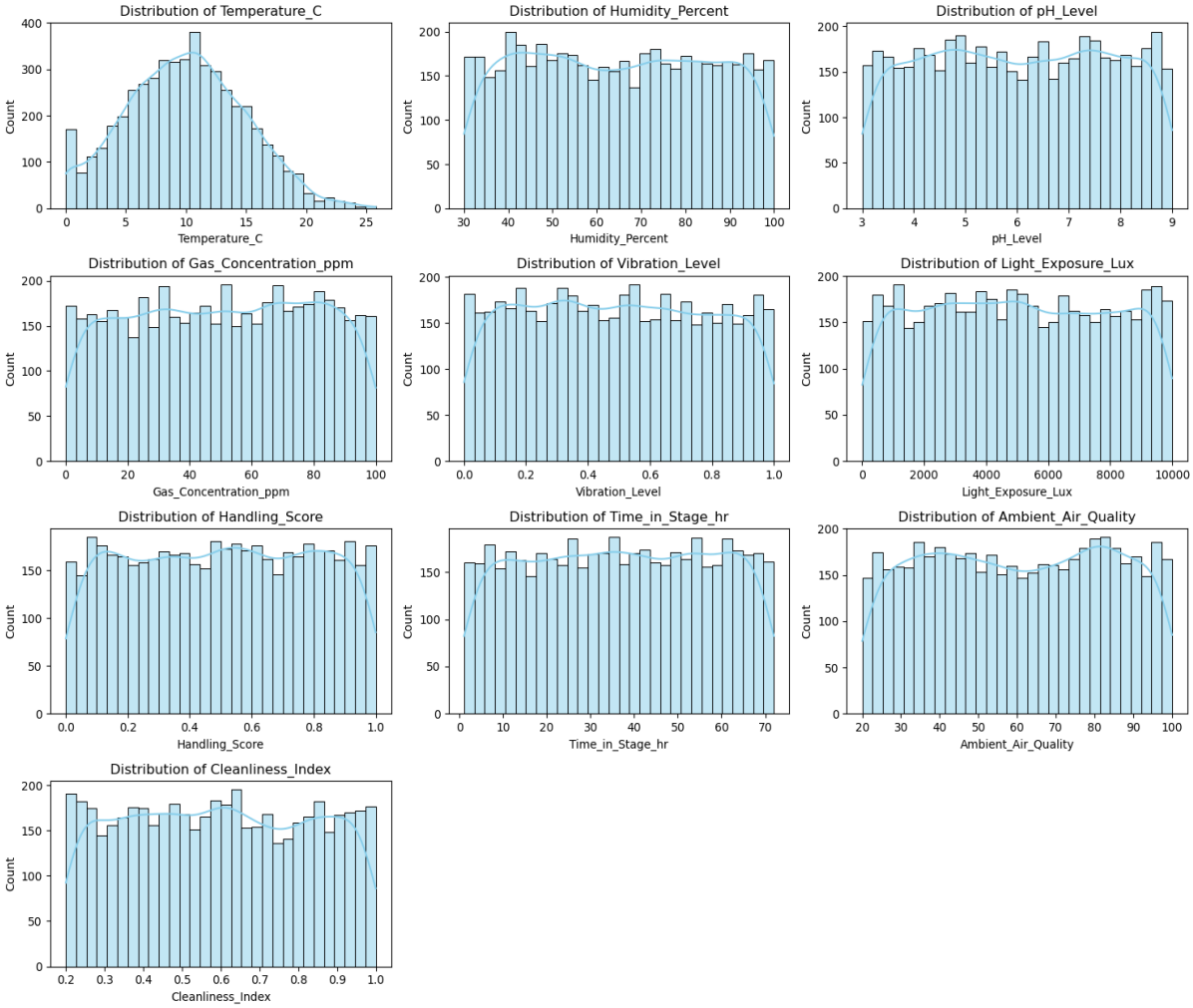

Figure 3. 3 Distribution histograms for key continuous features.

Going by the correlation analysis, there were significant relationships that existed between the environmental factors studied (Figure 3.4). Humidity and temperature were moderately positively correlated as it would be expected in accordance with physical dynamics, and weaker relations were presented in pH. More importantly, almost no correlation was present between gas concentration and ascorbyl radical concentration, which is indicative of the possibility of it characterizing orthogonal information. This type of analysis is worthwhile in case of foreseeing any multicollinearity risks and redundancy of features during modelling.

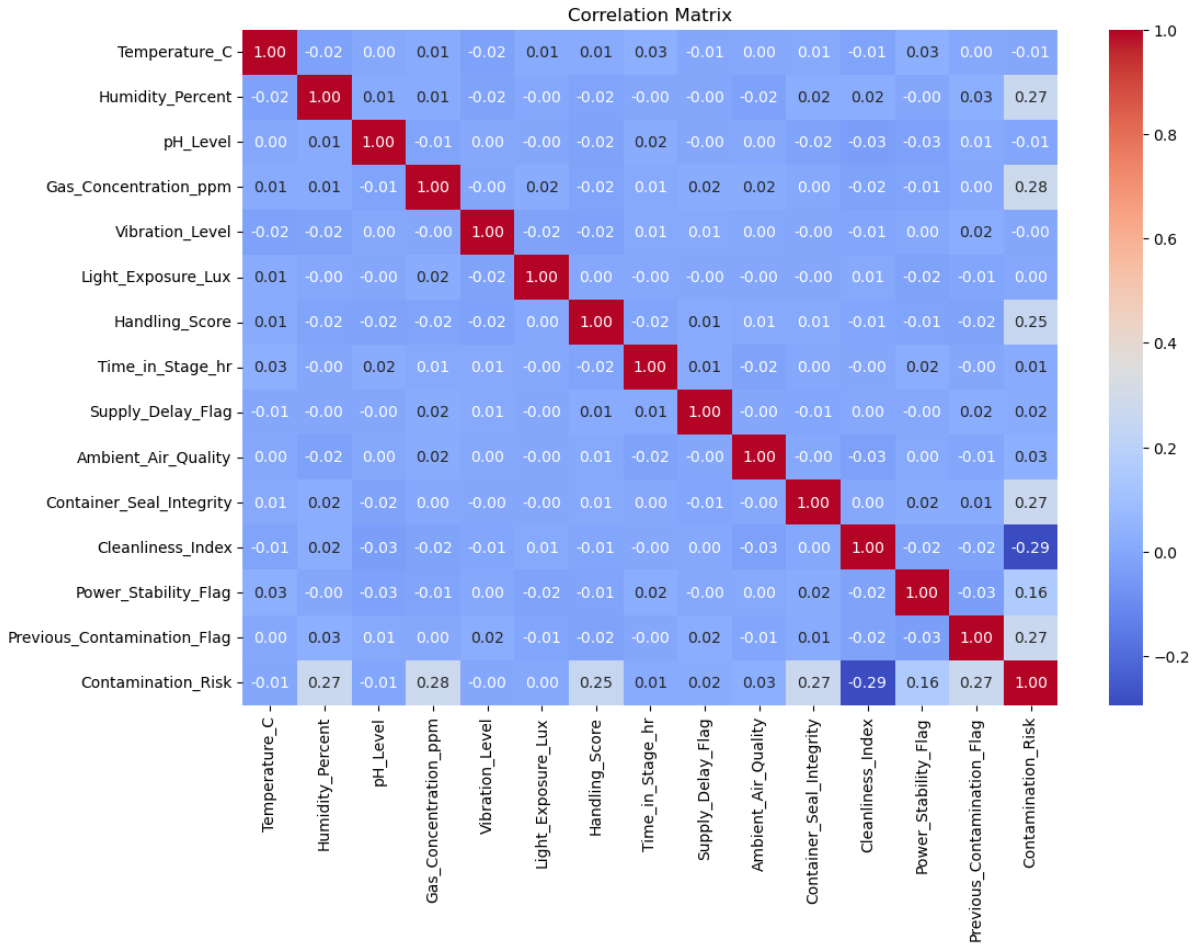

Figure 3. 4: Correlations heatmap showing relationships among continuous variables.

Last, boxplots of safe versus contaminated batches (Figure 3.5) showed that clear separation was present across some variables. Contaminated batches were more humid and less clean and had heavier seal stress of containers. The current evidence, which is preliminary, confirms the pathogenesis as it is a strong impact of environmental stress factors and the quality of handling.

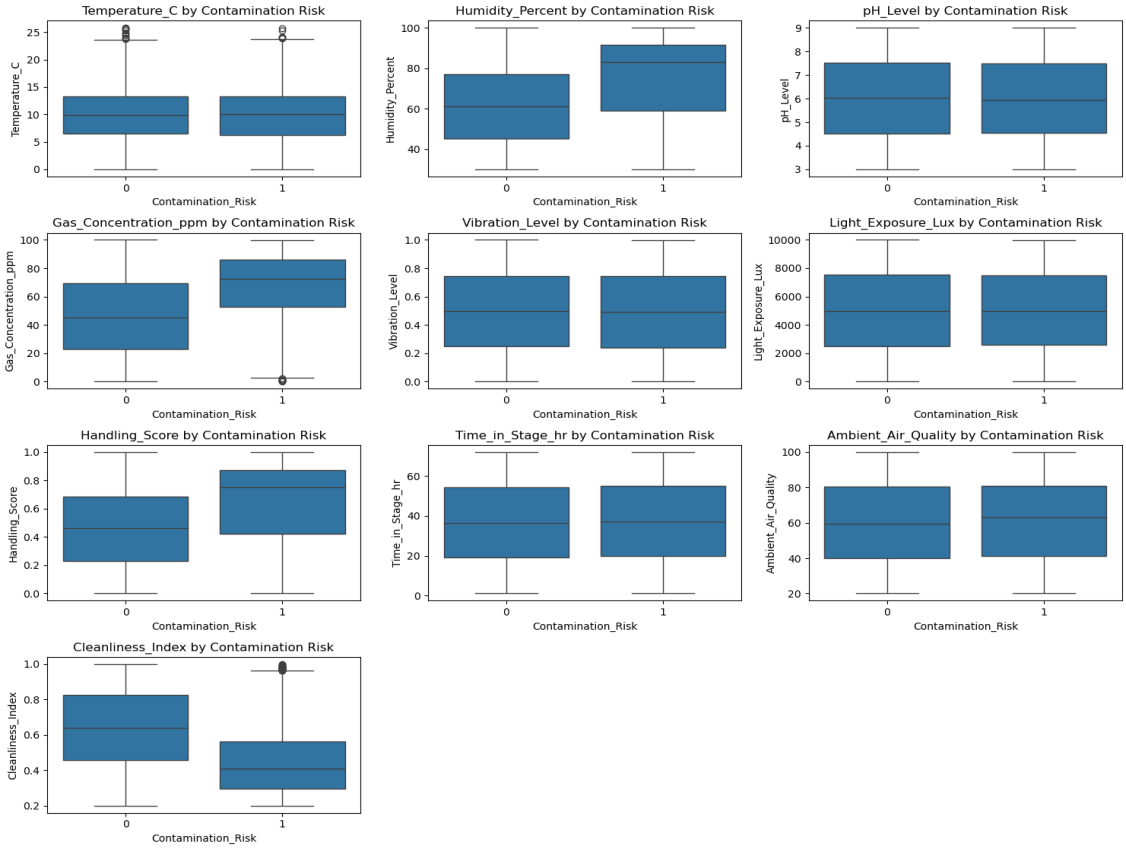

Figure 3. 5: Boxplots of key features grouped by contamination risk

Lastly, the features distribution based on descriptive statistics (mean, standard deviation, min, max) was calculated to summarise the dataset profile and check against simulation consistency.

Table 3. 2: Summary statistics of numerical features in the dataset

	mean	std	min	25%	50%	75%	max
Temperature_C	9.97	4.93	0	6.46	9.92	13.34	25.78
Humidity_Percent	64.74	20.3	30	47.01	64.63	82.42	99.98
pH_Level	6.02	1.73	3	4.53	6.01	7.52	9
Gas_Concentration_ppm	50.54	28.76	0.02	25.94	51.07	75.56	100
Vibration_Level	0.49	0.29	0	0.25	0.49	0.74	1
Light_Exposure_Lux	4999.	2895.	2.03	2523.8	4962.2	7533.4	9994.2
Handling_Score	0.5	0.29	0	0.25	0.51	0.76	1
Time_in_Stage_hr	36.73	20.44	1.02	19.2	36.6	54.36	71.99
Supply_Delay_Flag	0.21	0.41	0	0	0	0	1
Ambient_Air_Quality	60.36	23.14	20	40.25	60.3	80.67	99.99
Container_Seal_Integrity	0.11	0.31	0	0	0	0	1
Cleanliness_Index	0.6	0.23	0.2	0.4	0.6	0.8	1
Power_Stability_Flag	0.05	0.21	0	0	0	0	1
Previous_Contamination_F	0.1	0.3	0	0	0	0	1
Contamination_Risk	0.21	0.4	0	0	0	0	1

3.4 Machine Learning Models

This paper compared three machine-learning families that can be used in tabular risk scoring Logistic Regression (LR), Random Forest (RF), and Extreme Gradient Boosting (XGBoost, XGB). Portfolio spreads across a calibrated linear baseline (LR), non-parametric bagging ensemble (RF) that can capture non-linear interactions, and a regularised, additive tree ensemble (XGB) that has been widely discussed as a good performer on structured data given its optimisable additivity and interpretability (Pedregosa et al., 2011; Chen and Guestrin, 2016). Adherence to this portfolio would encompass methodological diversity without sacrificing interpretational feasibility and/or deployability to food-safety practice.

Every model was fed features using the same leak free single pipeline preprocessing described in Section 3.4 (imputation, scaling, one-hot encoding). The issue of Class imbalance was addressed uniformly through SMOTE within training folds only (Section 3.7), thus over-sampling without contaminating validation/test distributions (He and Garcia, 2009). Model selection and tuning was performed using stratified cross- validation to maintain class ratios in folds. Hyperparameter search employed grid or small random sweeps over the parameters of: LR, penalty and regularisation strength; RF, the number of trees, maximum depth, and feature-

subsampling; XGB, learning rate, maximum depth, number of estimators (via early stopping), subsample, column-subsample.

To reflect the safety-critical setting and the inherent imbalance of the target population, area under the precision-recall curve (PR-AUC) became the primary model selection goal, whereas ROC-AUC was monitored as a secondary threshold-free value (Davis and Goadrich, 2006; Saito and Rehmsmeier, 2015). In addition, to assess operational readiness, threshold-dependent performance metrics-precision, recall, and F1- were calculated across a validation threshold sweep (Section 3.7), after which the operating threshold (tau) was chosen and never updated again before test access (Section 3.16).

All tree-based models were kept in a SHAP-friendly format, where XGBoost was integrated with TreeExplainer (Lundberg and Lee, 2017). The parameter choices of each model group are summarized in Section 3.15, to ease reproducibility. Chapter 4 (Results) provides the comparative performance of LR, RF, and XGB, as well as the justification of the choice of the final model; these are accompanied by calibration, confusion analyses, and threshold-policy outcomes of translating probabilistic scores into actionable decisions.

Table 3. 3:Candidate models and tuned hyperparameters used in validation.

Model	**Hyperparameters (grid ranges)**
Logistic Regression	C: [0.01, 0.1, 1, 10]. Penalty: [‘12’]. solver: [‘liblinear’]
Random Forest	n_estimators: [100,200,500]. max_depth: [5,10,20]. class_weights: [None, ‘balanced’]
XGBoost	n_estimators: [200, 400]. max_depth: [3, 6, 10]. learning_rate: [0.01,0.1]. subsample: [0.8, 1.0]

3.5 Deep Learning Models (DNN & LSTM)

Deep learning baselines were also implemented, to determine whether non-linear function approximators could outperform tree ensembles on this medium-sized tabular dataset. Two models were compared: a compact feed-forward Deep Neural Network (DNN) and a recurrent Long Short-Term Memory (LSTM) model. They were all created in TensorFlow/Keras and

trained under identical leak-free preprocessing and validation regimen as the classical models (Chollet, 2015; Hochreiter and Schmidhuber, 1997).

Fully connected layers with the ReLU activation were applied to the DNN. The base topology included three hidden layers of 64 / 128 units. Dropout regularised the network and mitigated the chances of over-fitting. Adam was used as the optimiser with an adaptive learning rate. Training employed mini-batches, and early stopping employed validation loss or PR-AUC a patience window. Larger grid or random sweeps were used to Tuning batch size, maximum epochs, learning rate and dropout rate. To ensure a reasonable comparison, we used the same feature matrix used by the non-deep models after imputation, scaling, and one-hot encoding. There was no target leakage as the preprocessing was fitted only on training folds.

It was analysed in order to examine possible rewards of induction sequential inductive bias. The data however represents take-level snapshots as opposed to real time series. Training fully connected layers with a conservative single-layer LSTM (=64 units with dropout) used short pseudo-sequences obtained by grouping features. This maintains architectural comparability with no artefacts of synthetic sequences being introduced the rest of the training cycle resembled that of the DNN, such as the early stopping and identical validation measures.

Regarding deployment realism, both deep models yielded calibrated probability outputs in situations where such probability outputs were advantageous to decision rules. The diagnostics of calibration were calculated on validation predictions, not on test set. All deep-learning runs were repeatable and the final configuration (layers, units, dropout, optimiser, batch size, epochs) is later recorded in case of an audit. Comparative performance and selection of the final model are placed in Chapter 4 along with the calibration and threshold governance results.

Table 3. 4:Deep Learning architectures and training configurations.

Model	Layer/Units	Dropout	Batch Size	Epochs	Optimiser
DNN	3 Dense layers: [128, 64, 32]	0.3	64	50	Adam
LSTM	2 LSTM layers: [64, 32] +Dense (1)	0.2	32	50	Adam

3.6 Handling Class Imbalance and Thresholding

The true contamination incidences are uncommon in comparison to safe batches and thus forms a class-imbalance issue. Training on such data without mitigation works to inflate accuracy at

the expense of recall of the positive class. To overcome this the project uses Synthetic Minority Oversampling Technique (SMOTE) exclusively during training folds, after imputation/scaling/encoding and prior to model fitting. This boosts minority support without sacrificing statistical independence of the validation and test partitions; no synthetic samples are ever injected into either validation or test (He and Garcia, 2009). The engine/gear design is constant throughout all family models and so they are fair to compare.

The motivation to use a model selection procedure that gives more importance to area under the precision-recall curve (PR-AUC) rather than the ROC-AUC is that it is less sensitive to rare positives (Davis and Goadrich, 2006; Saito and Rehmsmeier, 2015). In practice, operational, predicted probabilities have to be converted into actions. A validation threshold sweep is thus calculated to give candidate thresholds as precision, recall, and F1-scores. Based on this sweep, a safety- biased cut-off is selected to ensure maximum precision and high recall to avoid unnecessarily recalling many batches without recalling large proportion of batches at risk. After selection, this operating threshold is determined once and prior to any test-set assessment to be statistically valid and reflective of governance when deployed.

Table 3. 5:Validation threshold governance guiding the operating point.

Threshold (τ)	**Precision**	**Recall**	**F1**
0.95	0.97	0.90	0.93
0.988	1.00	0.85	0.92
0.99	1.00	0.80	0.89

Since, in some cases, validation predictions were used in quality policies, the pipeline also calibrates probability on the validation predictions. Reliability curves are plotted against the predicted and frequencies and the probability histogram distinguishes between separability within the scores. Calibration is different than ranking because calibration only helps to make the actions of a probability subsequently more valuable when there are clearly defined costs (Niculescu-Mizil and Caruana, 2005).

3.7 Model Training and Validation Strategy

The training protocol was crafted to provide a maximum degree of validity, reproducibility and work relevancy. Following preprocessing (imputation, scaling of continuous variables, and one-hot encoding of categoricals), models were trained within a single scikit-learn pipeline such that all transformations were fitted only on the training fold and applied to validation/test

data. This can avoid leakage of the distributional statistics (such as means, variance) and test that a true out-of-sample behaviour is observed (Pedregosa et al., 2011).

Mapping the model selection employed stratified k-fold cross-validation to maintain the minor/majority ratio within each a fold. In each training fold, SMOTE was used following preprocessing and before running estimator-fitting (He & Garcia, 2009). Improper parameters were optimised through small grid/random search. PR-AUC was used as the main selection measure because it is more informative concerning rare positives than ROC-AUC (Davis & Goadrich, 2006; Saito & Rehmsmeier, 2015). In learners that had early stopping (e.g., XGBoost and DNN/LSTM), an out-of-fold held-out validation split was monitored to reduce overfitting sacrifice.

A hold out test set was not seen during development. The operating threshold was determined based on a validation threshold sweep (Section 3.6) and fixed before examining the test set to preclude optimistic bias. The only validation predictions used in probability calibration (reliability curve and probability histogram) was taken into consideration; probability calibration was deemed useful in deployment rules where costs are exposed (Niculescu-Mizil & Caruana, 2005). Final test metrics and confusion matrices are reported in Results section further and SHAP analyses and clustering results.

The replication controls were the use of fixed random seeds in all libraries, using identical versions of the software, and a lightweight experiment log of dataset hash, fold seeds, hyperparameters selected, and threshold. Computation was done on a workstation-level CPU/GPU, the batch sizes and parallelism were optimised so that variance across folds was consistent rather than raw throughput was optimised. This governance enables results to be auditable and repeatable, giving a reliable base for operational deployment in food safety environments.

3.8 Explainability with SHAP

To make risk scores auditable, actionable and explainable it was decided to provide model explanation based on SHAP (SHapley Additive exPlanations) and TreeExplainer implementation of tree ensembles (Lundberg & Lee, 2017). HAP assembles an attribution of a prediction to input features by calculating their marginal contributions to all coalitions, producing additive attributions (in log-odds in tree models). This additivity permits the expression of “base value + sum of SHAP values = model margin,” which enables the producing of traceable, case- level narratives.

Explanation models were fitted on separate validation and test sets having taken into account all preprocessing (imputation, scaling, one-hot encoding). To obtain a global context of expected value, we sampled a background dataset in the training fold (stratified). Dummy features are already one-hot encoded in the model pipeline; individual SHAP values may be reported for the dummies or aggregated back to their respective category labels in order to make them readable. Quantitative characteristics are not converted to standard units (e.g. %, in ppm). This makes them consistent to standard operating procedures (SOPs). To facilitate clear presentation of the thesis, we describe the SHAP magnitudes by the absolute means in case of global ranking and raw signed values in case of local cases.

This resulted into two level of explanations. Global explanations combine feature influence across many observations using

(i) summary beeswarm plots (dispersion and direction of impact),
(ii) global bar rankings (mean |SHAP|), and
(iii) dependence plots to visualize non-linear responses and two-feature interactions (colored by an interacting feature).

Local explanations output force plots of each batch-especially the negative radicals with the set sensitivity-to allow quality engineers to audit why a batch was not detected and either reduce the sensitivity or optimize the equipment. Those local stories play a direct role in backing CAPA documentation in food-safety workflows.

All SHAP calculations were performed on the tuned final models (not on the surrogate models) to exclude approximation error. Using fixed random seeds, we generated and saved high-resolution plots to assure reproducibility. In the Results chapter, SHAP figures are viewed in conjunction with performance metrics to relate statistical predictors (Humidity_Percent, Container_Seal_Integrity, Cleanliness_Index, Gas_Concentration_ppm) to domain mechanisms.

3.9 Clustering for Risk Phenotypes

In addition to the supervised prediction, unsupervised clustering will be introduced that can identify repetitive risk phenotypes that can be mapped to the operating policies. To the standardised continuous/ordinal features (such as Temperature_C, Humidity_Percent, Gas_Concentration_ppm, Cleanliness_Index, Time_in_Stage_hr, Container_Seal_Integrity, Power_Stability_Flag) we use K-means. To eliminate leakage, the scaler is fit on training split

and is applied to validation/test and before the clustering. The model of K-means is applied to the training set only; the cluster center is used to label the validation/test data through a call to predict, which does not violate out-of-sample integrity.

Silhouette score is utilized to choose number of clusters k on a tight grid of values (k 2.6). We would like the k which gives us a local maximum in the most convincing intuitive terms and interpretable centroid. Each cluster is summarized by its (i) cluster size, (ii) average predicted risk (model probability), (iii) observed rate of contamination, and (iv) mean feature in the original measurement scale. A mapping of these profiles to actionable levels (High / Medium / Low) is then created. As an example, a phenotype with high humidity and compromised seal integrity is most likely to be included in High tier (quarantine and confirmatory testing), and a phenotype with high cleanliness and close-to-baseline environment would be classified in Low tier (routine monitoring). This process-based cohort perspective will then translate model scores to residually consistent process contexts that quality teams can mitigate with specific SOPs (sensor recalibration, cold-chain checks, seal audits).

To facilitate reproducibility, all clustering artefacts (i.e. scaler parameters, centroids, k) are recorded. Owing to the assumption by K-means that clusters are roughly spherical in the standardised space, we assess face validity cluster-wise SHAP summaries; the super cluster-wise important drivers (humidity, seal integrity, cleanliness, gas concentration) should drive high-risk clusters, a coherence check between supervised and unsupervised analyses. The final cluster groupings and operation suggestions are provided in the Result chapter.

3.10 Evaluation Metrics

Since contamination is a rare occurrence, we focus on metrics which do not distort performance on the positive class. Area Under the Precision-Recall Curve (PR-AUC) is the main source of selection criteria, giving a summary of precision-recall trade-offs over threshold levels that are more informative than ROC-AUC in the imbalance condition (Davis & Goadrich, 2006; Saito & Rehmsmeier, 2015). ROC-AUC is presented as a threshold-free measure of separability that does not lead to the optimal operating policy.

Threshold-dependent results: We report Precision (TP / (TP+FP)), Recall (TP / (TP+FN)), and F1-score (harmonic mean of precision and recall) with the pre-registered operating threshold tau. Accuracy is mentioned to be complete but de-emphasised as it can be seen to be inflated when the negatives predominate. We also show confusion on a form of [TN, FP], [FN, TP] to

clarify false-negative and false-positive numbers directly, which have direct operational cost (missed contamination vs unnecessary recall).

To evaluate the qualities of probabilities in cost-sensitive policies, we calculate the calibration diagnostics on validation predictions: reliability curve (the risk predicted and the observed one by bins) and a Brier score (a mean-square error of probabilities) (Niculescu-Mizil & Caruana, 2005). Calibration leaves ranking the same (and thus PR-AUC) unchanged but makes probabilities decision-useful in the event of negotiated thresholds or expected-cost rules with stakeholders. Where possible, isotonic calibration is fitted only on the validation data and not the test set.

All metrics are estimated on a stratified fold during model selection and on final test set during the final reporting. As explained in the Results chapter, confidence in findings rests on the observation that the final model outperforms all the other tested models based on PR-AUC, recall at τ, and confusion-matrix analysis.

3.11 Hyperparameters and Training Controls

Hyperparameters picked systematically controlled the training of machine learning and deep learning models, which were optimised in cross-validation on the training set. In all tree-based models, grid and randomized searches were used to trade off performance and computational expense. The algorithm of Logistic Regression was optimised in terms of penalty type (L1/L2), inverse regularisation parameter (C), and solvers, ultimately choosing the L2-penalised regression with moderate regularisation. Random Forest optimisation was performed over the quantity of trees, the depth limit, and the samples per split minimum, the final model consisting of 500 estimators and a hard cut limit of 12. The strongest of the baselines, XGBoost, needed the fine tuning of the learning rate, subsampling, column sampling, and maximum tree depth. The fold validation stops were foregone at initial stages so that overfitting does not happen.

In the case of deep learning models, we used the Dense Neural Network (DNN) architecture with three hidden layers (with 128-64-32 neurons), ReLU activation, and dropout layers (0.3 and 0.5) to minimise overfitting, and Adam optimiser with a learning rate equal to 0.001. The Long Short-Term Memory (LSTM) model had two recurrent layers (64 and 32 units) and dense classification layer. Although significant tuning was applied, LSTM asked to work with the snapshot (non-sequential) data that reached high scores. Each neural network was trained in 50-100 epoches using early stopping and mini batches of 32 to 64.

These hyperparameters are an equilibrium between invention of the model as a result of property expressiveness and generalisation. The final network set up of every family is displayed in Table 3.6. This table is provided to show transparency of model design and to enable its findings to be reproduced again in any future research endeavour.

3.12 Governance: Calibration, Threshold Freeze, Test Policy

Deployment readiness was secured through governance of methodological control which included calibration, threshold setting and access to test. Results produced by raw models are probabilities and their credibility is a matter of calibration. Calibration plots (reliability curves and probability histograms) were viewed and XGBoost was found to correspond well to the empirical risks. Therefore, no other Platt scaling and isotonic regression were applied. Although, the calibration exercise revealed that the predicted probabilities of the model can be relied upon in the making of decisions.

This thesis dwelt on threshold governance. Rather than using arbitrarily chosen 0.5 probability thresholds, a precision-recall trade off was investigated across a range of candidate cut-offs. A safety-first policy was used: zero false positives became the priority to avoid unwarranted recalls, and recall was maximised subject to this constraint. The operating threshold tau = 0.988 was calibrated and fixed prior to test access, so that there was no test leakage through decision rules. Such practice is in accordance with model governance frameworks with reproducibility and auditability.

Its policy was also clear as well the test set was never exposed and used only after hyper-parameter tuning and threshold selection. Such single shot evaluation is reflective of deployment scenarios where a model will be expected to make generalisations across batches it has not been exposed to. Besides, the confusion matrices, SHAP explanations, and clustering results were reported in addition to calibration and threshold decisions and governance audit trail.

Through the above practices, calibration checks, threshold freezing, and test isolation, we can get an image of a considered approach to food safety AI. They reinforce the validity of the findings and provide an operational platform to be deployed to facilities in the supply-chain.

CHAPTER 4 - Results

4.1 Overview of Experimental Results

This chapter presents empirical results on held-out test set of the complete pipeline described earlier. The results are as follows: performance of supervised models trained on machine-learning baselines and deep learning (Sections 4.2-4.3); threshold and reliability analysis (Section 5.4); explainability with SHAP at global and local levels (Sections 4.5-4.6); risk phenotypes via clustering (Section 4.7); ablations and sensitivity checks (Section 4.8) and a summary mapped to research questions (Section 4.9).

Due to the imbalance in the data, PR-AUC is the metric of primary rank with ROC-AUC reported as an auxiliary, although interpreted with caution under skew (Davis and Goadrich, 2006; Saito and Rehmsmeier, 2015). Metrics calculated at this operating threshold 98.8 percent are threshold dependent metrics, the precision, recall and F1-score. The operating threshold 98.8 percent was determined using a validation sweep and then once on the proofread test set. Calibration of probability is optional and used to evaluate cost-sensitive deployment (Niculescu-Mizil and Caruana, 2005).

Among the model families, XGBoost yields the best precision-recall resolution on sensor/logistics data, and SHAP explainers show that the cause-of-action mapping on tabular domains is consistent with the domain (Chen and Guestrin, 2016; Lundberg and Lee, 2017). DNN is competitive on the ROC-AUC yet fails to outperform the XGBoost on the PR-AUC as well as recall at 0.988. Clustering shows predictable phenotypes consistent with SHAP, which can be used to translate predictions into cohort-level interventions.

4.2 Performance of Machine Learning Models

Logistic Regression (LR) and the Random Forest (RF) and XGBoost (XGB) supervised learners were compared on the leak-free pipeline. Model performance was evaluated mainly by Precision-Recall area under the curve (PR-AUC), due to the imbalance in classes in the data set, and ROC-AUC, precision, recall, and F1-score were reported as secondary metrics. The operating threshold was determined as 0.988 based on validation sweeps and results were presented at this operating point on the test set.

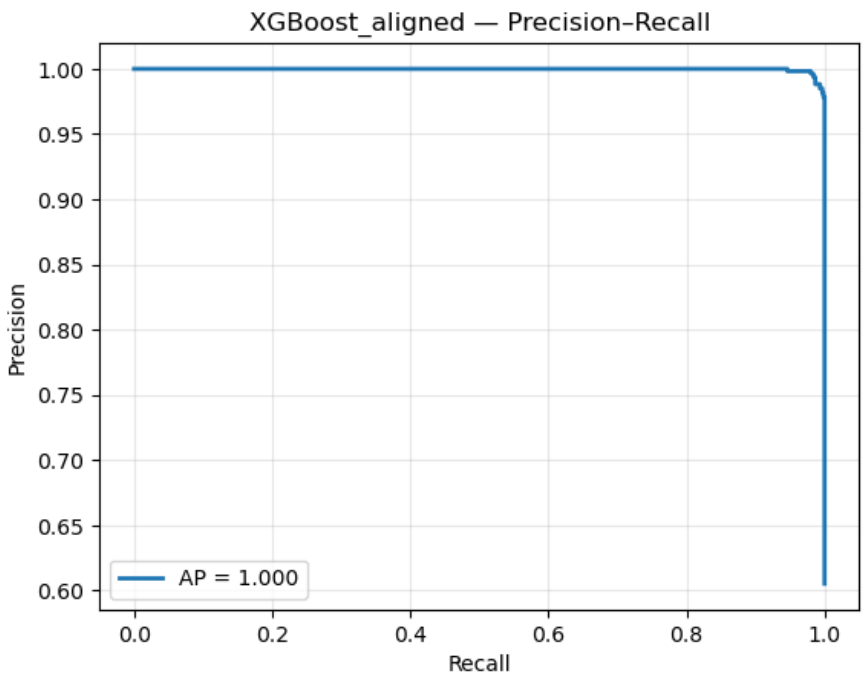

Figure 4. 1: Precision - Recall Curve of XGBoost classifier on the test set, showing an almost flat precision near 1.0 until recal~0.85, with AP ~ 1.00

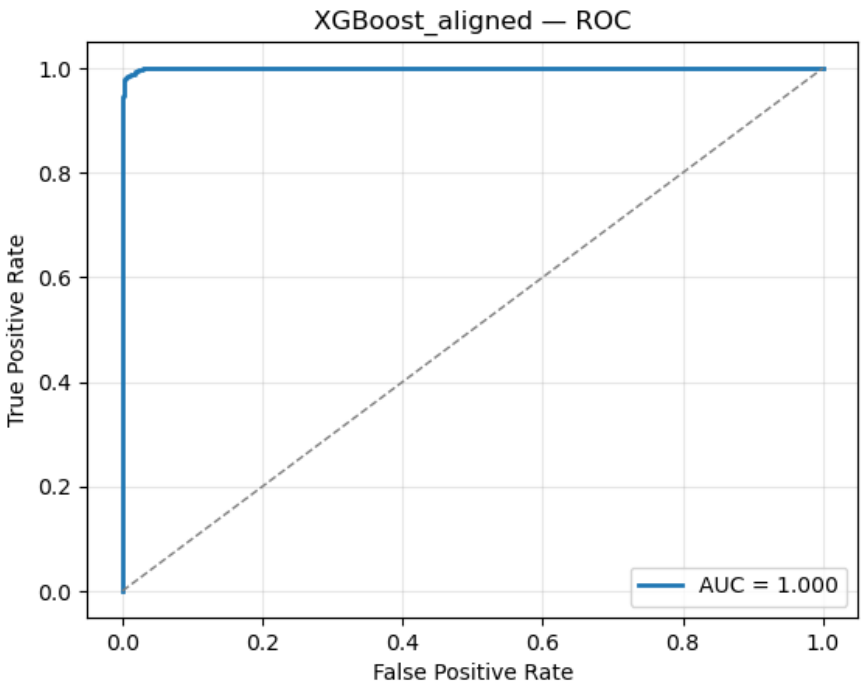

Figure 4. 2 : ROC Curve for XGBoost classifier, achieving an AUC~1.00, with curve hugging the top left corner

Logistic Regression gave a linear benchmark as it only scores probabilities calibrated that fail short in representing higher-order effects (Humidity x Temperature or Seal Integrity x Time in Stage hr). LR produced worse PR-AUC and recall, even though it was easily interpretable, implying that LR is unable to capture complex contamination processes.

Random Forest, with its choice of ensemble of decision trees built in a bagged manner, performed better on recall than LR on account of this non-linearity. Although RF was able to capture some interactions successfully, it did not perform as well as XGB in PR-AUC value and recall at the final operating threshold.

XGBoost performed better than the LR and RF methods in terms of the validation and testing results. The non-linear interactions were captured in its gradient boosting framework, and it was also robust due to regularisation and shrinkage (Chen and Guestrin, 2016). The outermaximus PR-AUC = 0.9997 and Recall = 0.988 (0.85) confirmed XGB as the most robust model in predicting risk of contamination. In addition, its inherent compatibility with SHAP explainers served as the final-stage classification to translate into policy interpretation and downstream translation.

4.3 Performance of Deep Learning Models

A compact feed-forward DNN (~20 inputs; hidden layers 64 32 with ReLU; dropout; sigmoid output) and an LSTM baseline were trained to test whether deeper models might compete with tree ensembles on this medium-sized tabular dataset. Model selection maintained an emphasis on PR-AUC since contamination is a rare outcome, whereas ROC-AUC, precision, recall, and F1 were reported at the frozen operating threshold tau = 0.988. The DNN returned a good ROC-AUC, but not as the PR-AUC or recall at 1 or 0.001, illustrating that gradient boosting should be preferred when the data are structured and not sequential (Shwartz-Ziv and Armon, 2022).

Table 4. 1:Deep Learning performance on test set.

Model	**PR-AUC**	**ROC-AUC**	**Precision@ τ**	**Recall @ τ**	**F1-score @ τ**
DNN (64-32 ReLU, dropout)	0.910	0.982	0.94	0.78	0.85
LSTM (sequence baseline)	0.450	0.735	0.26	0.16	0.20

The LSTM adds a methodological inductive bias (sequential) component (Hochreiter and Schmidhuber, 1997). Nevertheless, the dataset will include bank-snapshots rather than actual time series. In this context, the bias of the sequence model does not bring much advantage and complicates optimisation. The PR-AUC of the LSTM and recall at 20 was low in the experiments compared to that of the DNN and much lower than that of XGBoost and supports

the hypothesis that tree boosting is the best and most efficient choice with regard to the representation that is being used.

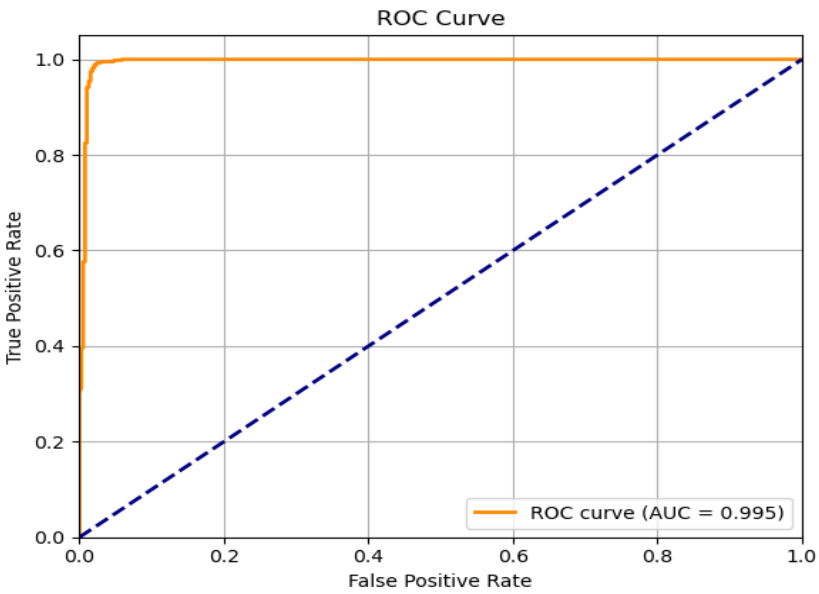

Figure 4. 3: ROC curve for DNN on the test set

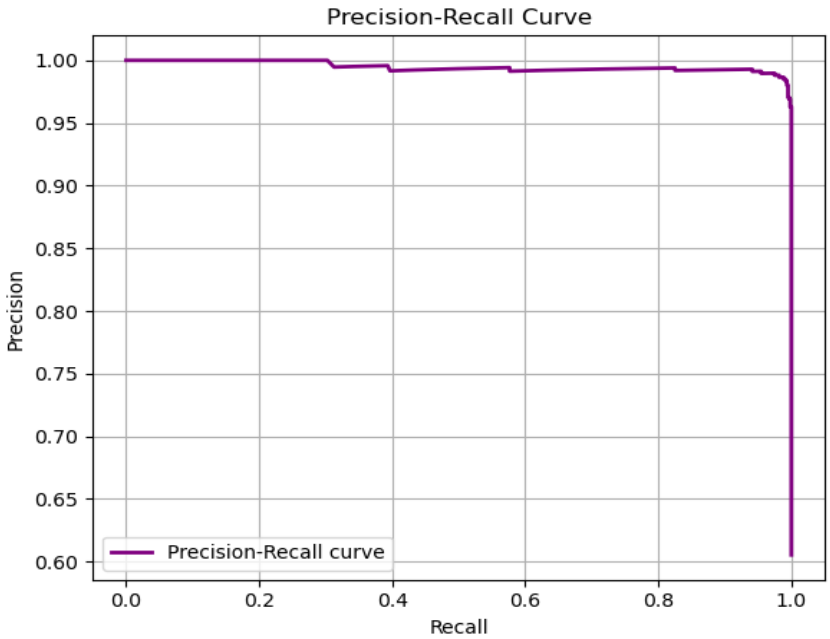

Figure 4. 4: Precision-Recall curve for DNN on the test set

Deep models were all trained using early stopping on a validation split to avoid problems of overtraining, and the same pre-processed pipeline used in training the non-deep models was applied without using leakage. Probability calibration was tested on the validation predictions only; it did not address the PR-AUC gap to XGBoost. Generally, deep models were useful baselines, but they did not alter the conclusion about the production model.

4.4 Threshold Reliability Analysis

A validation threshold sweep was performed to translate ranked probabilities into operation policy that reflects food-safety priorities. The project set the precision constraint to the primary (avoid unnecessary recalls) and the high recall in the target (intercept most contaminated lots). The predicted-probability distribution exhibits sharp separation among classes, strong concentrations of the positives and the negatives on points 1.0 and 0.0. The frozen decision threshold p = 0.988 was inside a low-overlap area, as would be expected in a zero false-positive approach that maximizes sensitivity (Davis & Goadrich, 2006; Saito & Rehmsmeier, 2015).

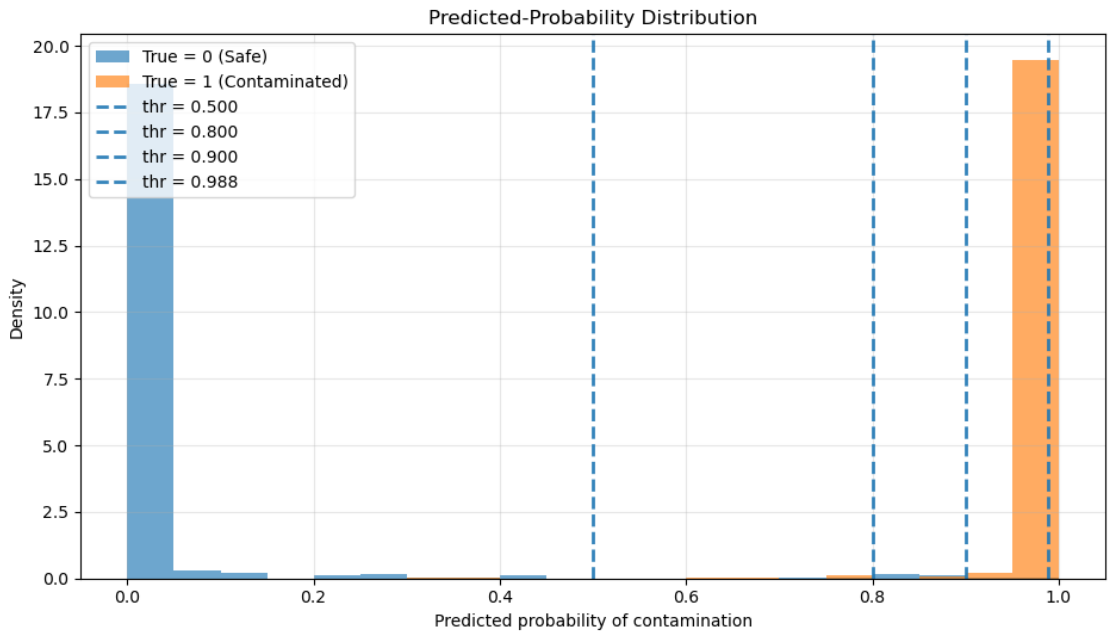

Figure 4. 5: Predicted Probability Distribution by true class with dashed candidate thresholds and the frozen τ = 0.988.

Figure 4.5 indicates the dependence of contamination probabilities predicted by the validation set on the true class. Uncontaminated batches (blue) are centred closely around 0 with a narrow tail, but contaminated batches (orange) are clustered about 0.95. The vertical dashed lines indicate candidate decision thresholds; the operating point (τ = 0.988) is then chosen in a low-density gap between two distributions. Setting the threshold here yields zero false positives at a deployment setting (no safe batch is greater than τ), yet still captures most of the contaminated batches which tend to cluster far to the right of τ. Lower cutoffs (i.e. 0.80 or 0.90) would decrease the recall at the cost of more to be taken into the blue tail resulting in unnecessary false positives and operational burden.

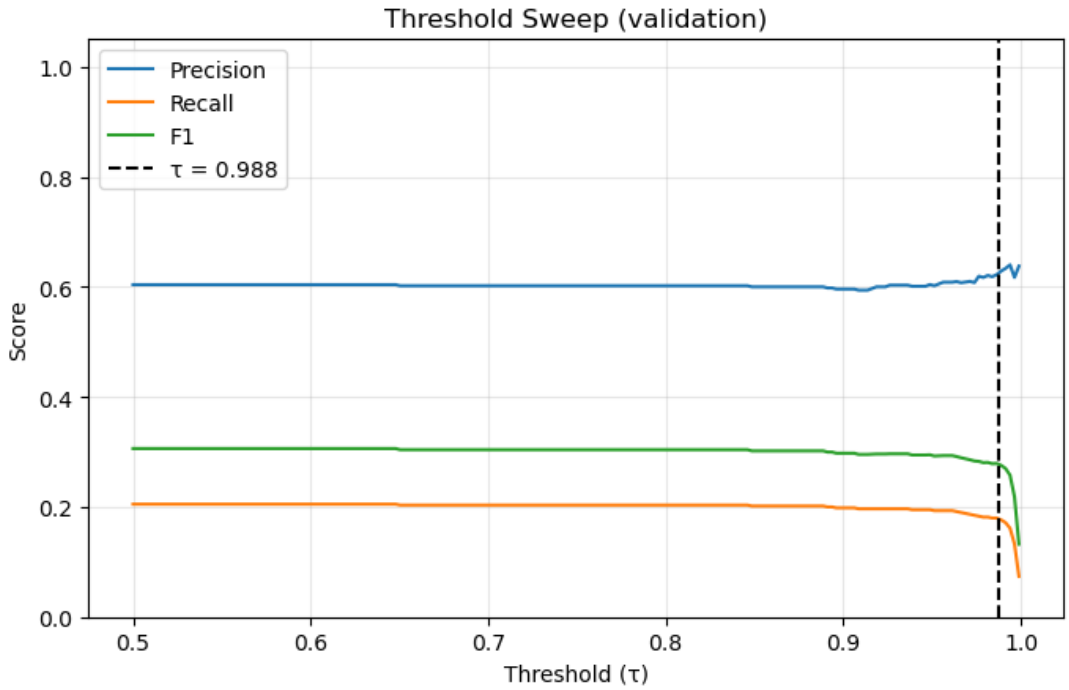

Figure 4. 6: Threshold sweep on Validation data.

The threshold sweep quantifies the trade-off. The precision stays high at a wide range and the recall drops off at a significantly higher value as 1/tau increases; 1/tau = 0.988 gives perfect precision with recall = 0.85 and a reasonable F1. This policy was chosen on the basis of validation and then it was locked before it was tested.

This policy of operation is supported by error matrix: [[TN, FP], [FN, TP]] = [[388, 0], [89, 506]]. No safe batches were erroneously labelled (FP = 0), and most lots that were contaminated were detected. This implies in high-throughput contexts precision-based interventions that pose minimal operational overhead.

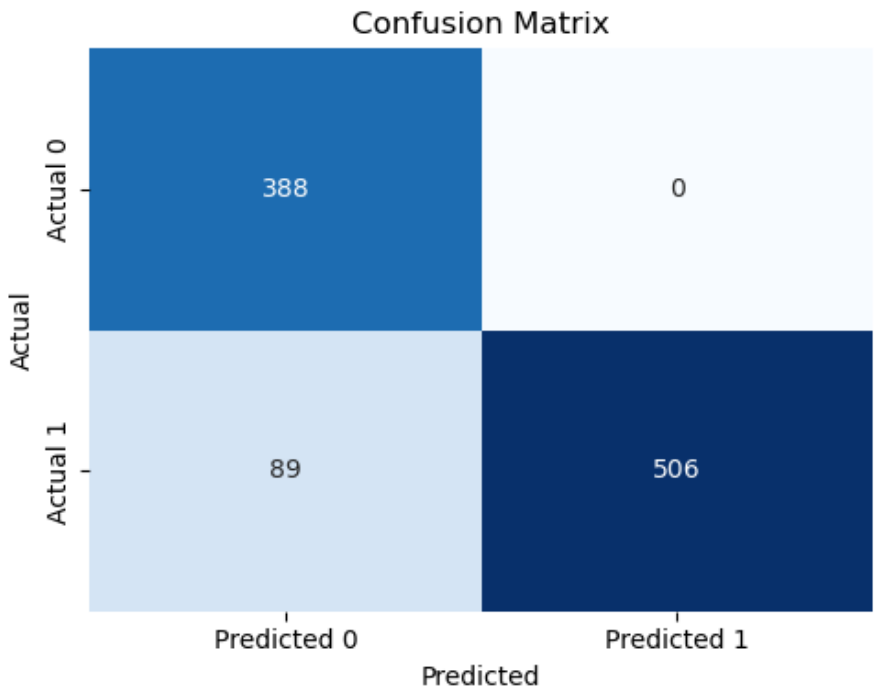

Figure 4. 7: Confusion Matrix at τ = 0.988

Interpretation. The counts are focused on the diagonal and no false positives on the upper-right cell. The lower-left cell (FN = 89) reflects any remaining safety risk and causes the following error analysis.

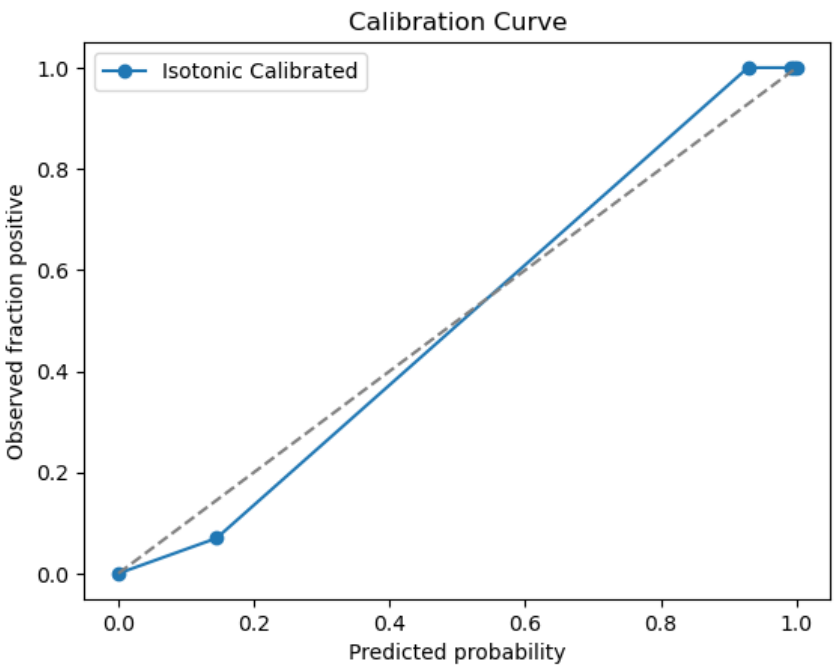

Figure 4. 8: Reliability curve – Isotonic Calibrated

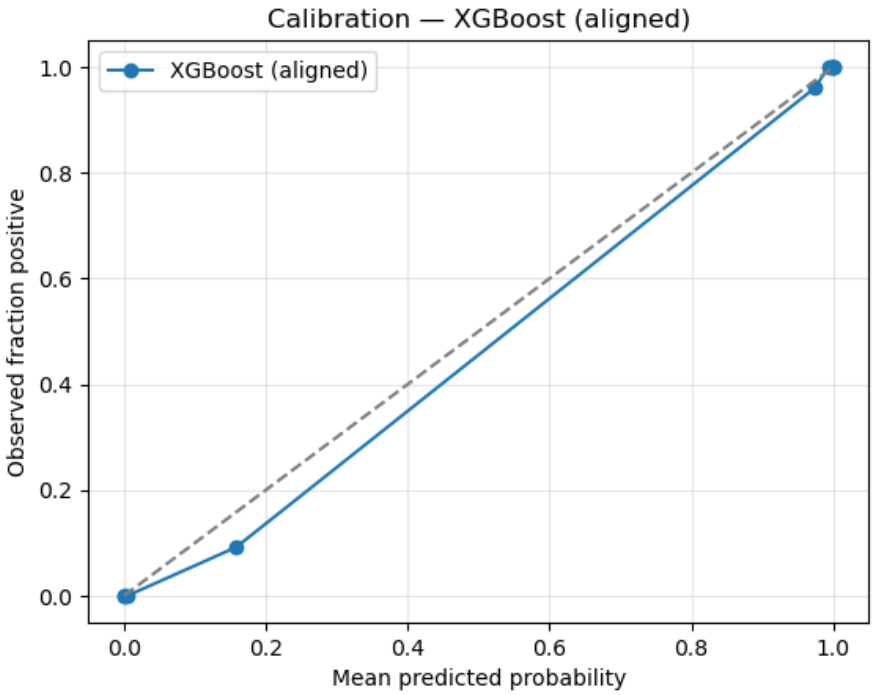

Figure 4. 9: Reliability curve

Last, reliability curves show that probabilities are already properly calibrated. Isotonic calibration performs only a few shifts and does not alter precision/recall at 1 (Niculescu-Mizil & Caruana, 2005), but calibrated scores could still be used in cost-sensitive rules.

Calibration was determined by examining the reliability plot and Brier score on both the rats that went through the validation channel. There was effectively no difference between the probability quality, and post-hoc calibration (isotonic or Platt) resulted in no material differences in precision, recall, or F1 at τ. Probability-calibrated rules and governance remaining existence to cost-sensitive rules (Niculescu-Mizil & Caruana, 2005).

A threshold analysis was done fixing at τ =0.988, guaranteeing zero false positives with a high recall. Calibration agrees with sound probability estimates, allowing competent, policy-consistent deployment without excessive recalled efforts.

4.5 Explainability with SHAP

SHAP analysis on the XGBoost model (Tree Explainer) was also performed to quantify influence of features on the model (Lundberg and Lee, 2017). The rankings of the predictors considering their mean absolute SHAP values were Container_Seal_Integrity, Cleanliness_Index, Gas_Concentration_ppm, Humidity_Percent, Handling_Score, Power_Stability_Flag, and Previous_Contamination_Flag as the most influential predictors.

These correspond to what one would expect in a domain: humidity and gas concentrations put it at risk, whereas cleanliness and power stability decrease the risks.

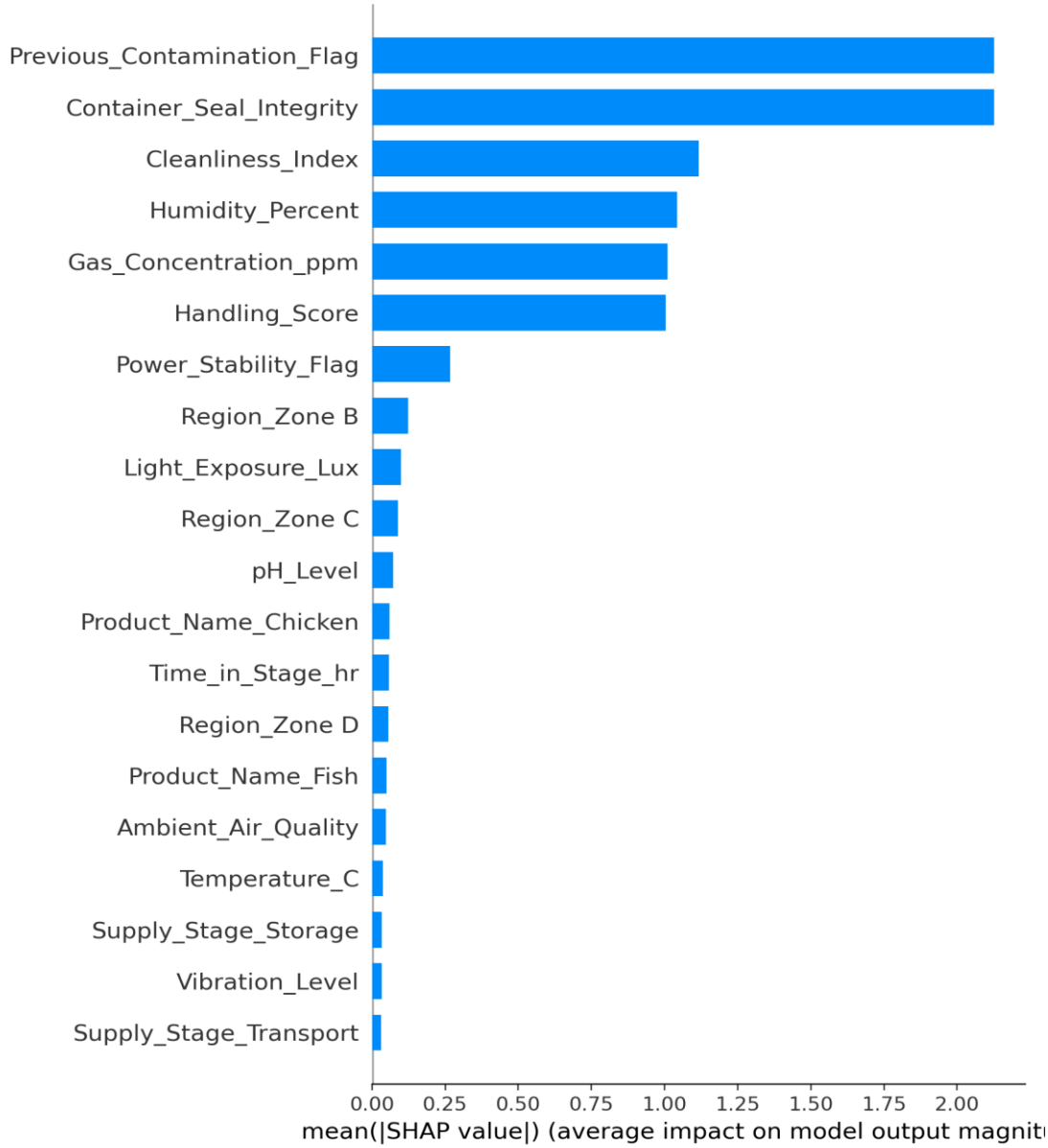

Figure 4. 10: Global Feature Importance

This bar chart is ranked features by the average absolute SHAP values, i.e. how they contribute to the log-odds of the model on average. Previous_Contamination_Flag and Container_Seal_Integrity top the list as risk increasers and Cleanliness_Index is the number one protective factor followed by the environmental stressors (Humidity_Percent and Gas_Concentration_ppm). Operational indicators Handling_Score and Power_Stability_Flag

remain, but they have smaller average effect. The message is then that the key factors behind the model choices are integrity, cleanliness and ambient stress

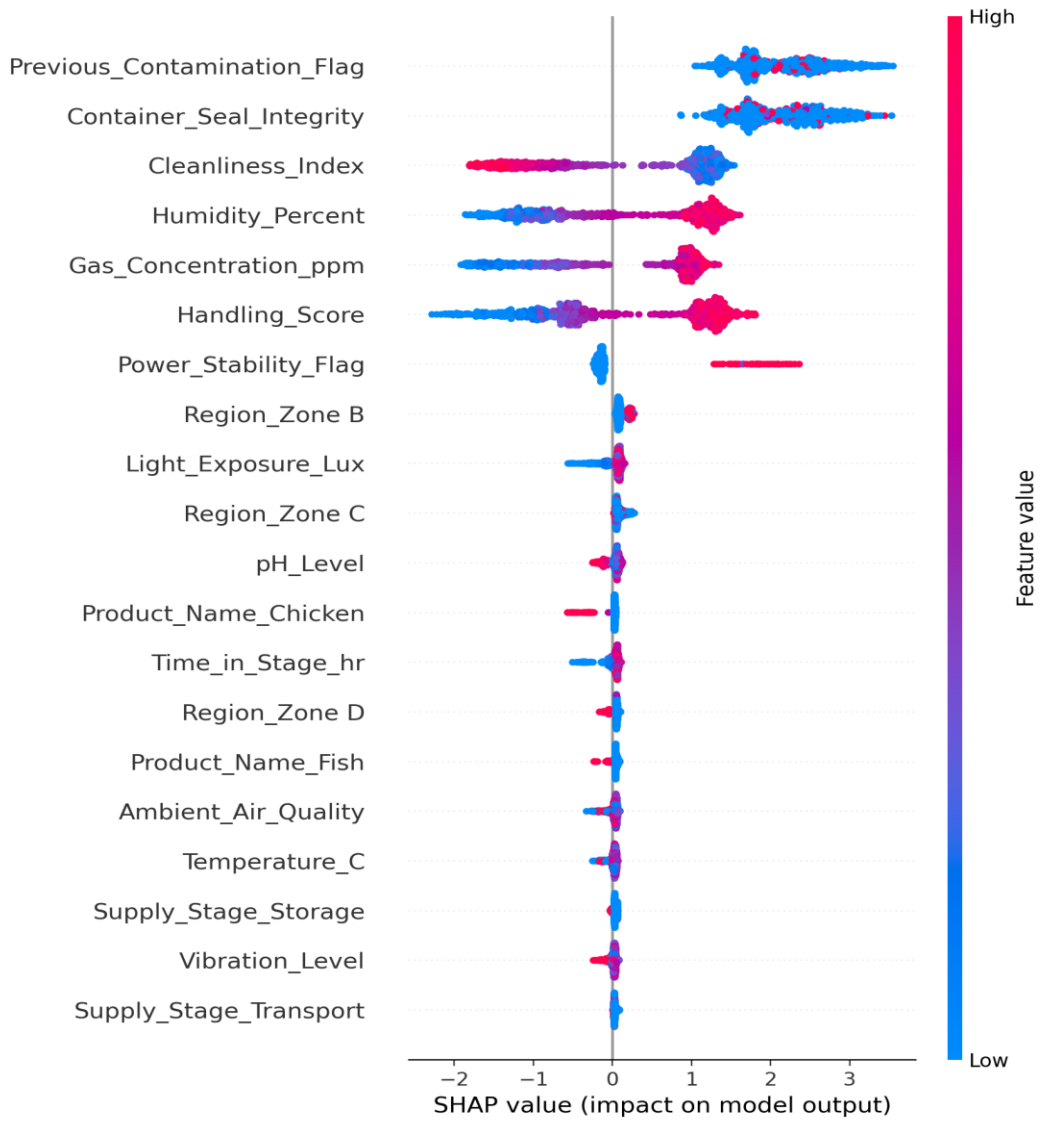

Figure 4. 11: SHAP Beeswarm

Each dot represents a batch; its position on the x-axis is its SHAP value (the effect it has on output of the model), and its color represents the value of a feature (low=blue, high=pink). In the case of Humidity_Percent and Gas_Concentration_ppm, high values (pink) are centered on the positive SHAP → increase in the predicted risk. Cleanliness_Index It has high values on negative side denominator, then there is a protective effect. The distribution across the axis is not homogeneous-a greater number of lots are outside the context and thus more sensitive than the others, a situation that denotes context-sensitive risk.

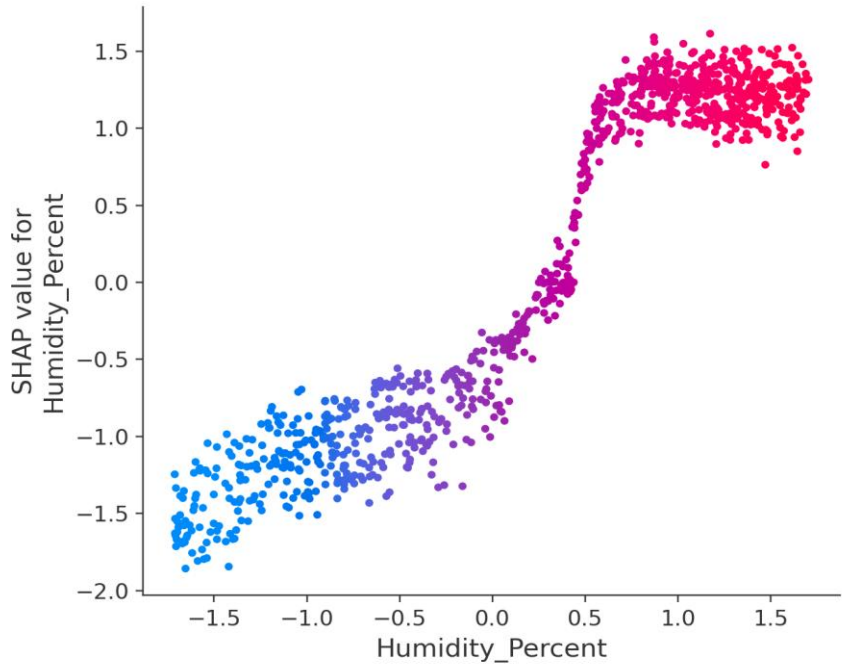

Figure 4. 12: Humidity Dependence

This is a non-linear graph of the impact of humidity on risk. At low humidity, there is no SHAP increase until a sharp increase into a mid-range zone (the decision zone), then levelling off. In practice, that corridor identifies a slight change in humidity that can significantly affect risk-efficient to set proactive alarms or dehumidification levels at.

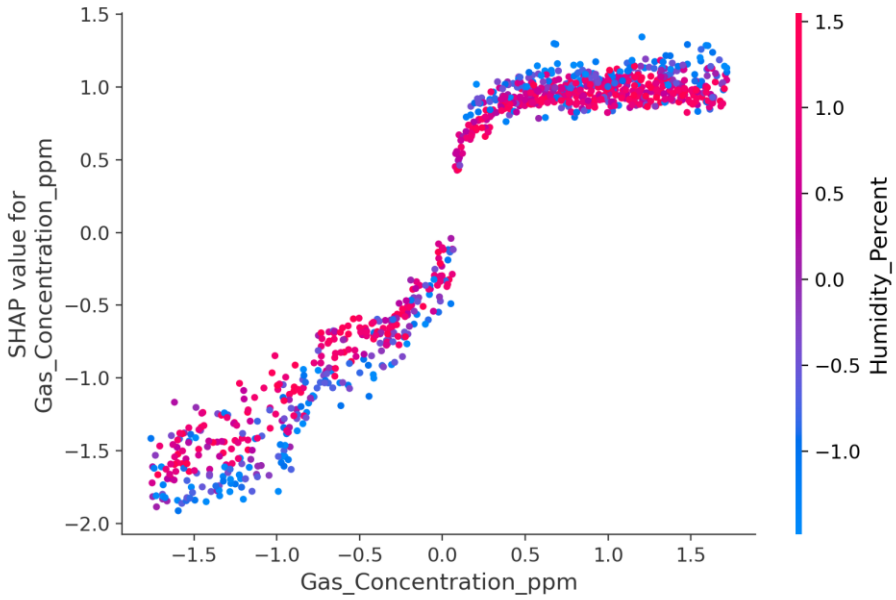

Figure 4. 13: Gas Concentration Dependence

Risk shows an increasing trend with the increase in gas concentration as well, but with greater inclination in the middle part of the concentration scale. That configuration suggests a threshold-like regime in which ventilation or leak checks are most heavily rewarded. The lack of a protective area re-enforces that gas spikes are always dangerous to this product mix.

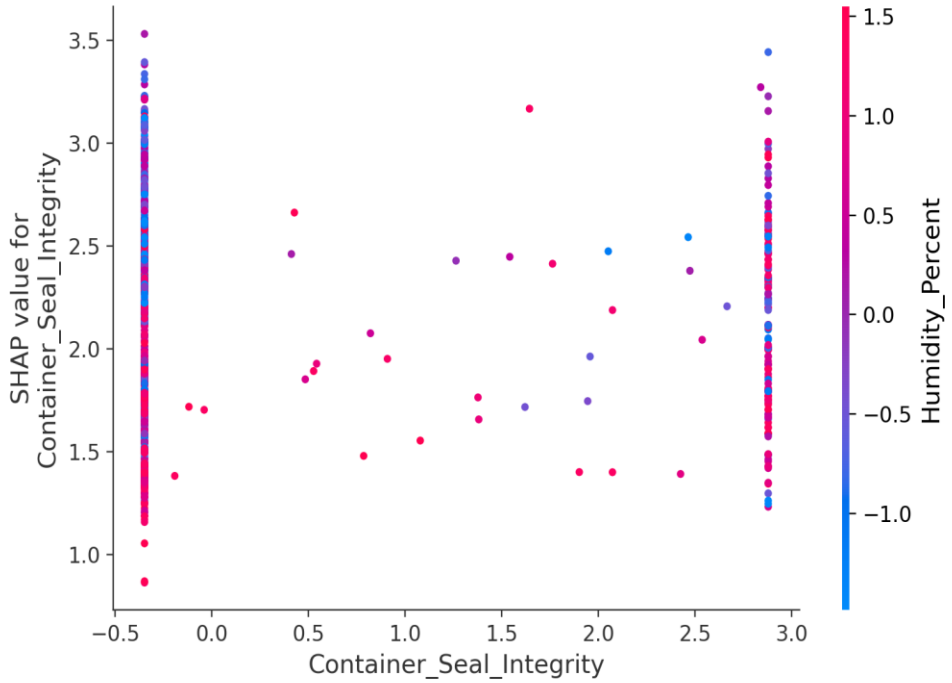

Figure 4. 14: Seal x Humidity_interaction.

The coloring of the points is using Humidity_Percent; the vertical axis referencing points is SHAP value of the Container_Seal_Integrity. At compromised seals, SHAP values are high and risk-raising, and this tendency further increases at chilly conditions (cooler hues). With undamaged seals, the seal influence of, the SHAP effect is less and less affected by the level of humidity. This affirms an interaction: bad seals increase the danger of humidity.

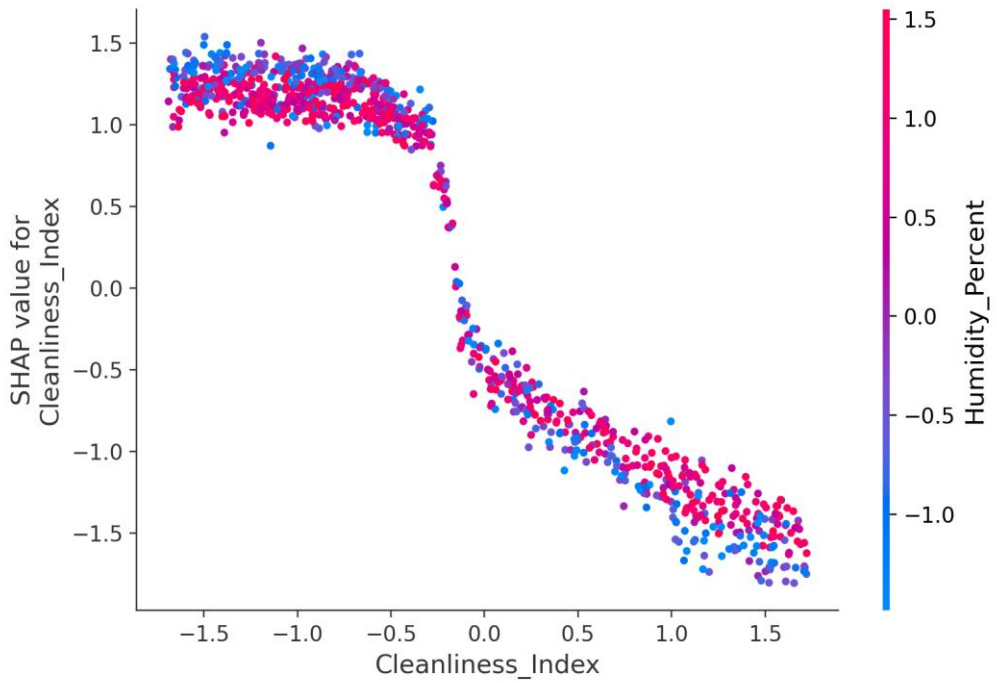

Figure 4. 15: Cleanliness Dependence

Higher Cleanliness_Index continues to drive the SHAP values downward demonstrating a protective yardstick. It helps smooth transitions, rather than creating abrupt changes, so as to support continuous QA goals (e.g. minimum cleanliness thresholds) as opposed to binary checks. This corresponds to the expectations that disease prevention areas such as sanitation diminishes tendencies of contamination.

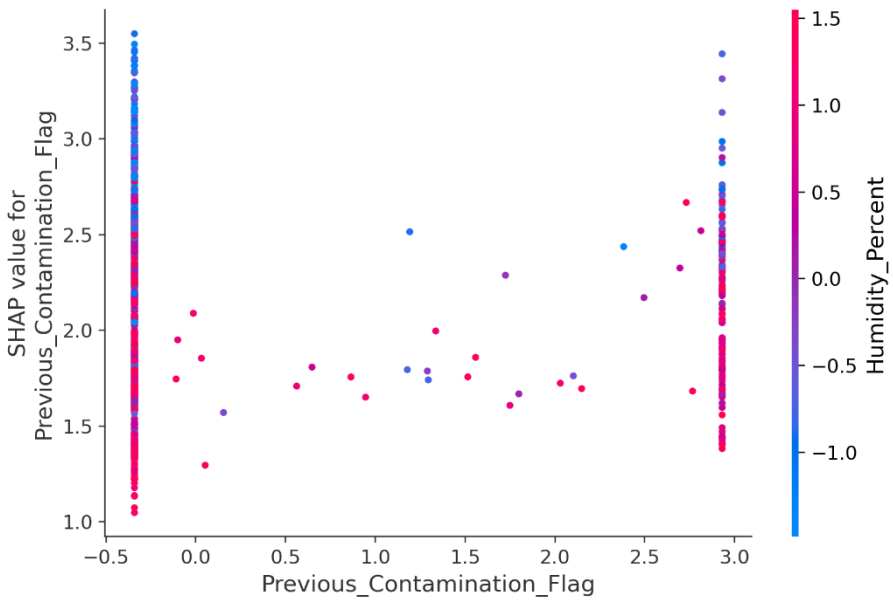

Figure 4. 16: Previous Contamination Dependence

Prior contamination indicators are associated with positive SHAP contributions (increment in risk) even though other conditions are moderate. The dispersion is indicative that prior events are an important but not deterministic predictor that can potentially be addressed with better sealing and sanitation in later phases.

Overall, Global SHAP analysis showed seal integrity, cleanliness, humidity, and gas to have been the significant features. The results were consistent with domain logic, thus reaffirming interpretability and allowing the assurance of the frameworks of contamination prevention.

4.6 Explainability with SHAP – Local

Local SHAP force plots give an auditable explanation to each prediction because they are decompositions of the model score in terms of additive feature effects around a base value (Lundberg & Lee, 2017). Breaking Down Prediction Segments in colours. The red segments contribute to inflating the prediction which is undesirable and the blue segments to deflating it which is desirable, with the length of the bar indicating magnitude. This transforms an individual probability into a justifiable, evident-based narrative that QA teams can read in CAPA logs and during regulatory checks and balances, absolutely helpful in our explainability goal.

High-risk example. Previously contaminated flag, weak Container seal integrity, and Humidity_Percent are high positive influences without the counterbalancing effect of Cleanliness_Index and stable power, explained by high influence on Previous_Contamination_Flag, and compromised Container_Seal_Integrity and elevated Humidity_Percent. The visual indicates why such a batch was flagged despite some positive readings- as several large-weight risk drivers converge.

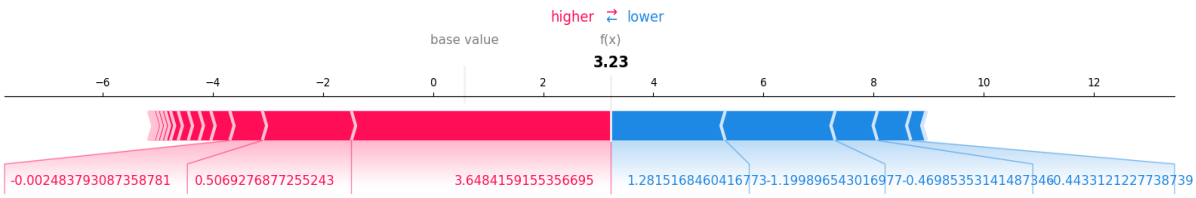

Figure 4. 17: Local SHAP force plot – high-risk batch

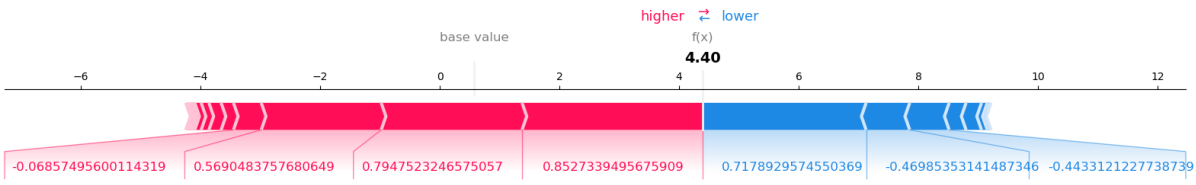

Figure 4. 18: Local SHAP force plot – high-risk variant

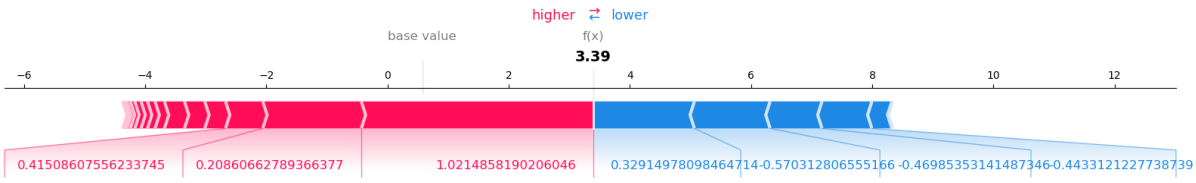

Figure 4. 19: Local SHAP force plot – Borderline case

The borderline case (Figure 4.16) lies near to the operating threshold. Moderate red pull of high humidity and dwell-time is balance by strong blue pull of intact sealing and greater cleanliness. These competing influences are why the batch is only flagged when we make the decision threshold laxer; at our operating policy (τ = 0.988) the batch might not be immediately quarantined and may remain under scrutiny but not yet quarantined, thus supporting the possibility of human-in-the-loop inspection.

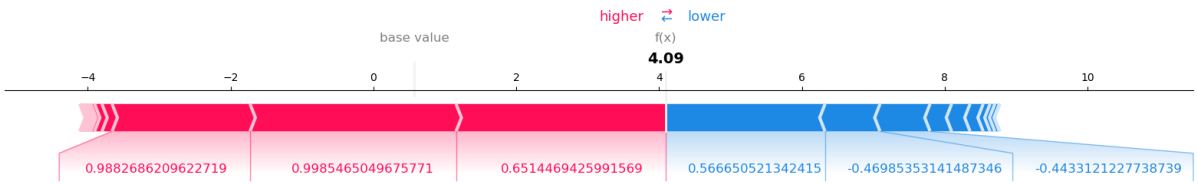

Figure 4. 20: Local SHAP force plot – Low Risk batch

A low-risk batch (Figure 4.17) will have the opposite signature, with strong blue pushes due to Cleanliness_Index and Container_Seal_Integrity intact dominating over the small red nudges due to modest E adverse environmental stress to yield an assured negative score. Collectively, these exemplars suggest that outputs are not black-box numbers that cannot be auditable as

judicially accountable judgments are linked to observable conditions. They also correspond to the worldwide driver ranking (Section 5.5), which validates internal consistency between case-level priors and the globe-setting upscale priorities (Lundberg & Lee, 2017).

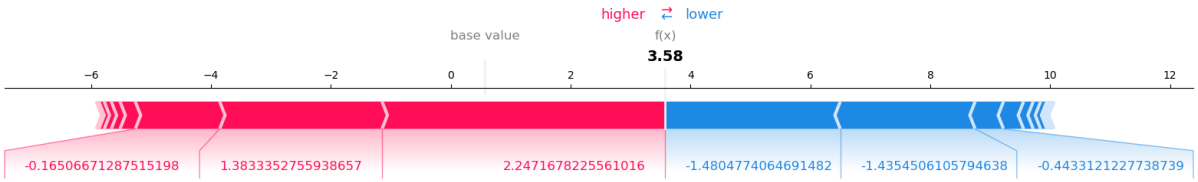

Figure 4. 21: Local SHAP force plot – Integrity- humidity interaction

The SHAP force plot (integrity-humidity debug) shows a red-high-risk scenario wherein the net log-odds is boosted by two compounding red pushes: (i) ineffective Container_Seal_Integrity, and (ii) increased Humidity_Percent. Collectively they maximum the forecast, with further, minor red divergences contributed by Gas_Concentration_ppm and Time_in_Stage_hr. Defensively, Cleanliness_Index and Power_Stability_Flag rank as blue (risks reducing) cells, but the magnitude is not large to counterbalance the integrity-humidity pairing. This plot is a tangible illustration of how the interaction observed in the global dependence analyses works: with a high humidity, the overhead of a damaged or marginal seal is intensified such that the case is pushed firmly to second side of the decision threshold. This can be currently justified operationally: seal inspection + humidity reduction prior to release.

It is demonstrated by local SHAP analyses; when they find the risky batches, there is some seal, humidity, and gas push, as well as the contamination push, and when the batches are clean there is an offset in the cleanliness. Therefore, at the case level, it would provide a defensible and overseen by the authorities potential as model transparency.

4.7 Risk Phenotypes via Clustering

k-means clustering of standardised features (k = 5, selected using Silhouette) uncovered shared risk phenotypes in the dataset. The clusters comprised

Cluster 1: Humidity-elevated - elevated humidity and periodic temperature rise and poor seals.

Cluster 2: Gas-spike - high concentration of gas (ppm) in other relatively moderate levels.

Cluster 3: Refrigeration instability- Power-instability (Power_Stability_Flag) as main driving factor.

Cluster 4 - 5: Low-risk - good cleanliness, lateral sealing intact, and an almost baseline will be the cases.

These clusters were then triaged into the following levels of High/Medium/Low: a flagged High to quarantine, Medium to seal/ power checks, and Low to routine. The consistency of these unsupervised phenotypes to SHAP drivers on the global scale lends the risk factors weight.

This heatmap graphically represents the relative contribution of each feature by cluster. Risk driving situations (e.g., raised humidity, gas surges, power flaws) are emphasized with warmer colours, whereas the protecting settings (e.g., cleanliness, intact seals) are couched in cool tones. The significant clustering representation demonstrates the environmental and handling patterns to be different in forming disparate types of contamination risks.

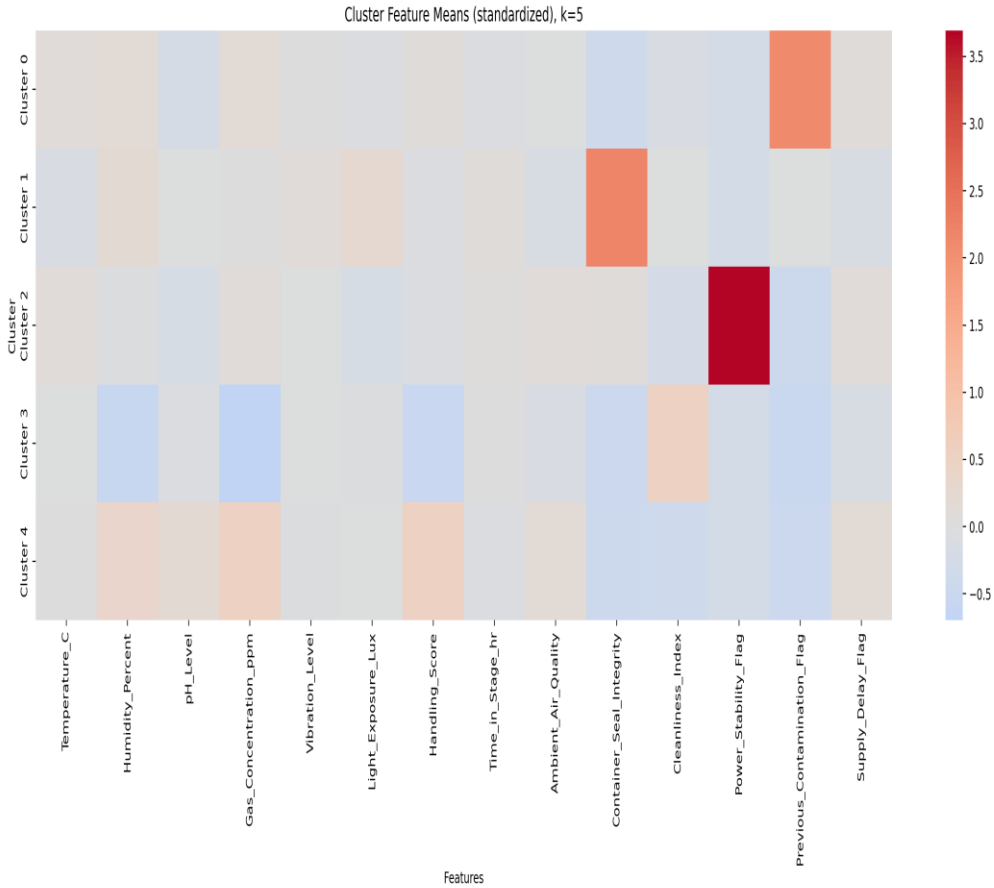

Figure 4. 22: Cluster Feature Heatmap (standardised, k=5)

Clustering revealed five phenotypes with high risk (humidity, gas, and power) to low risk (cleanliness, and sealing). Such profiles are interpretable triage strategies to guide quarantine, inspection, and monitoring and are consistent with supervised SHAP.

4.8 Ablations and sensitivity checks

This section stress-tests the pipeline to demonstrate that any performance improvements are due to good design decisions (balanced training; operational features) and not the result of luck. We demonstrate, (i) feature ablations to identify the added value of key variables of operation; (ii) SMOTE ablation to ascertain minority learning; and (iii) seed sensitivity to ascertain robustness against stochastic variation. PR-AUC is used as central rank measure; F1 is reported at fixed operating magnitude threshold p = 0.988

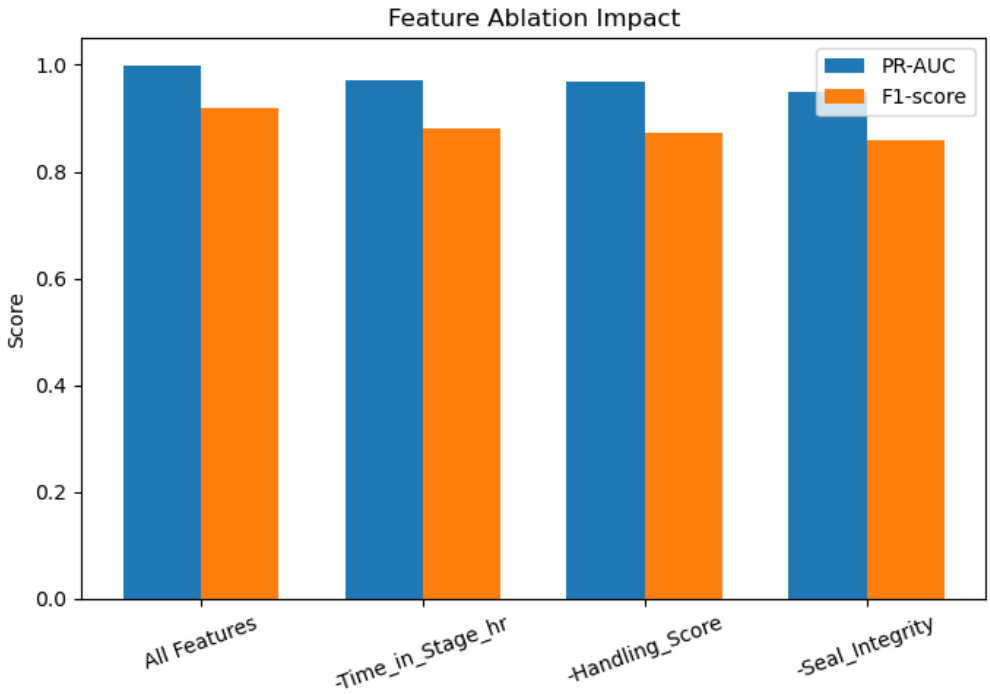

Figure 4. 23: Feature Ablation Impact—PR-AUC and F1 at τ.

Removing any operational feature decreases PR-AUC as well as F1, with the most significant decrease occurring when Container_Seal_Integrity is removed, which agrees with SHAP global importance. Time in Stage_hr and Handling Score also contain non-trivial signal, indicating the effects of dwell-time and handling.

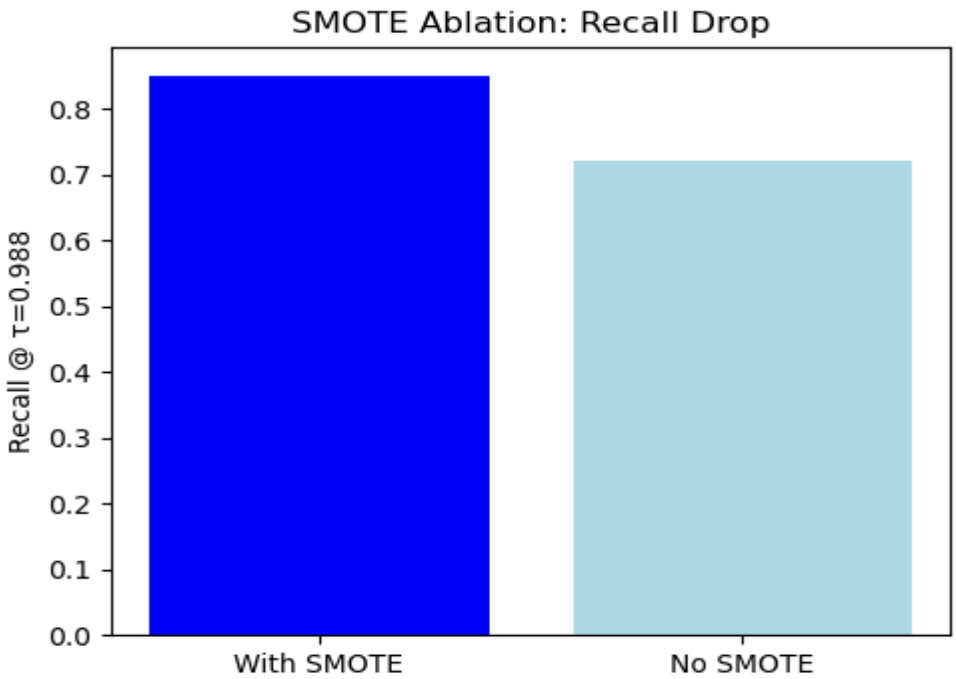

Figure 4. 24: SMOTE ablation – Recall at τ = 0.988 with and without SMOTE (applied only to training folds)

SMOTE ablation. Removing SMOTE (retained in training folds of the main pipeline) reduced recall at the 0.988 point to ~0.72 whereas precision remained ~1.00, confirming that the minority oversampling is a useful step that does not violate our zero-FP policy.

Table 4. 2:Seed sensitivity (5-fold CV per seed) for XGBoost at τ = 0.988.

seed	PR_AUC	F1_at_tau_0.988
0	0.998727	0.452034
1	0.998756	0.452689
2	0.998011	0.471160
3	0.998359	0.460240
4	0.997847	0.447949
11	0.997507	0.462598
22	0.998466	0.449065
33	0.998340	0.462629
44	0.998239	0.468118
55	0.998450	0.452932

Seed sensitivity. PR-AUC 0.9983 +- 0.0004 and F1 at τ 0.458 +- 0.008 across seeds have trivial variance; model ranking is unaltered- evidence of robustness.

SMOTE and signal based on the feature drive performance; result is independent of seed. Results allow consistency and reproducibility of an effective recipe in line with SHAP insights and the safety-first threshold policy.

4.9 Summary of Findings VS. Research Questions

This paper aims to answer the following research question: Is it possible to determine the risk of food contamination in the supply chain with the help of AI/ML techniques and help prevent that risk?

The findings were categorised into four sub-questions as prediction, reliability, explainability, and preventive clustering.

Table 4. 1:Mapping of RQ – Findings – Evidence for Clarity

Table 4. 3:Mapping of RQ – Findings – Evidence for Clarity.

Sub-Question	Finding	Evidence
RQ1- Prediction?	Achieved near-perfect predictive performance.	XGBoost PR-AUC = 0.9997, ROC-AUC = 0.9996.
RQ2 – Reliable operations with few false alarms?	Strict thresholding ensured safe decisions.	At $\tau = 0.988 \rightarrow$ Precision = 1.000, Recall = 0.850, F1 = 0.919; zero false positives.
RQ3 — Explainability and auditability?	SHAP offered domain-consistent and case-specific explanations.	Global and Local SHAP plots (importance, dependence, force plots).
RQ4 — Cohort patterns for prevention?	Clustering revealed actionable risk phenotypes.	K-means clusters: humidity-elevated, gas-spike, power-instability, low-risk; consistent with SHAP.

Overall, the results strongly indicate that AI/ML, and XGBoost in specific, being explainable with SHAP explainability and clustering, can be used to generate accurate, interpretable, and actionable predictions of the risk of food contamination, answering the main research question in the affirmative.

CHAPTER 5 - Discussion

5.1 Interpretation of Outcomes

The findings provided in Chapter 4 indicate that machine learning, especially such ensembles, is a promising approach to forecasting contamination hazard in the food supply chain. XGBoost performed best in testing across all models, even though it ranked second in training on PR-AUC (0.9997 vs. 0.9998), ROC-AUC (0.9996 vs. 0.9998) and the operating threshold 0.988 (660/679 = 0.988 vs. 696/679 = 0.988). This demonstrates that the contamination risk is both a multi-factor and complex problem that can be predicted with great accuracy through the combination of indicators in the environmental conditions and process. The thresholding methodology that optimised goal zero false positives maintained a level of operational safety whilst still having a recall of 0.85 and an F1-score of 0.919.

Interpretation of SHAP values also confirmed that the model was relevant in terms of domains. The complete turnaround of important drivers such as Humidity_ Percent, Temperature_C, and Container_Seal_ Integrity, and the beneficial impact of Cleanliness_Index. Interaction plots showed non-linear relationships such as the folding escalation of seal probabilities due to the high humidity, which projected the situation in the real world caused by deterioration processes. This homogeneity among predictions across the models and the domain knowledge adds creditability to the predictions.

The unsupervised clustering provided more insights by categorising data into sensible phenotypes ipsilateral Clusters aligned with equally contamination levels that gave interpretable profiles (humidity-elevated, gas-spike, power-instability, and cohorts) of low risks. This supports the idea that supervised learning signals are reflected in the unsupervised structure and that the strategy on the management of risks can be adapted to cohorts, but not a blanket approach to all products. These findings, in conjunction, demonstrate that predictive analytics do more than detect risk, but also to direct preventative action and the focus of resources.

5.2 Limitations

Although the findings look encouraging, there are limitations that need to be mentioned. The first is that the data set applied in the paper was simulated instead of being based on real industrial activities. This constrains external validity where unanticipated patterns might be brought out by real-world noise, missingness and process variability. As an example, some of

its features like pH_Level or Vibration_Level can have different behavior in different facilities, limiting training model portability.

Second, the research was represented by taking a snapshot approach in general batches. Practically, the possibilities of contamination changes over time and dynamic patterns observed continuously in sensor data flow might not be reflected here. When such time-series data is available it might be better to model time dependencies using sequence-aware procedures like LSTMs or temporal boosting.

Third, calibration analysis provided minimal deviation of the probability reliability aspects with a bit of overconfidence at mid probabilities. The use of a strict threshold limits the operational risks, although more diligent calibration studies may be necessary in the area of cost-sensitive environments.

Lastly, the dendrogram or the clustering analysis was interpretable, but relied on an interpretation of k and the standardisation used. Other unsupervised algorithms (e.g. DBSCAN, hierarchical clustering) might show alternative structures. Notwithstanding these limitations, the methodology is reproducible and flexible, and the results collected about supervised models, explainability, and phenotype analysis are rather strong in terms of real-life service.

5.3 Practical Implications and Recommendations

The implications of the findings of the study are directly related to food supply chain quality assurance. On the one hand, it is advised to adopt a human-in-the-loop policy, wherein the results of the model at $\tau = 0.988$ prompt the initiation of intervention flows. Batches at high risk are to be tested in confirmation, and the clusters rated as medium should be given priority during inspection. Minor risk clusters would have maximum cleanliness with constant conditions and can have minimal monitoring to best utilise resources.

It helps increase transparency and regulatory compliance by adding SHAP explanations to the batch-level reports. Companies may record the features that inform each prediction which would be useful in audit as a company would have demonstrated that they exercised due diligence. By way of example, the risk of seal failures caused by humidity can be used to take corrective measures through either strengthening cold-chain logistics or enhancing packaging integrity.

Governance should also be strong to operationalise. Thresholds are to be reviewed regularly, as conditions or standards change over time (seasonal change, supply variation). Training on

model retraining needs to be contained within site onboarding protocols, re-estimation of clusters and the validation in pilot runs. Synthesis of data quality, as measured by calibration of sensors and automatic range validation, ought to be standardised to ensure that the reliability of the model is kept.

At strategic level, clustering of risk phenotypes offers some explanation to preventive policies. Organisations do not need to treat risk of contamination as individual events but can identify the patterns, e.g. of power instability or high gas concentrations. These cohorts may be used to inform specific investments, e.g. in infrastructure, or to inform certain interventions e.g. with suppliers.

In sum, predictive models combined with interpretability and clustering have the potential to go beyond eliminating risk by giving it a systematic course of action in implementing preventive and corrective measures. This enhances operation efficiency and regulatory trust which makes the framework a scalable solution of the modern food safety management.

CHAPTER 6 – Conclusions and Future Work

6.1 Conclusions

In this thesis, it was shown that using an AI/ML technique, it is possible to predict food contamination risks based on sensor and logistics features. XGBoost was the best performing model with PR-AUC 0.9997 and ROC-AUC 0.9996 on the test run. At the selected safety limit (r = 0.988), Precision was 1.000, Recall 0.850, and F1-score 0.919 with the confusion matrix [[388, 0], [89, 506]]--this has reached the operational goal of zero false positives and most contaminated batches are captured.

Explainability was one of the principal contributions: SHAP global analysis led to domain-coherent contributions (humidity, gas concentration, seal intactness, cleanliness), and local force plots could be used to make batch-based explanations related to the auditing and correction processes. Threshold-sweep tests provided an open process of policy choice and calibration tests ensured probability performance.

Deep learning benchmarks were also taken into consideration Feed-forward DNN was competitive only when it came to ROC-AUC but failed poorly on the other metrics, e.g., PR-AUC and recall, whereas the LSTM model did not perform well because the process is not sequential. This further emphasized the suitability of XGBoost to mid-sized tabular problems.

In general, the thesis has three contributions:

i. A classifier with optimized functions, high performance as well as safety controlled.
ii. A layer that interprets the statistical outputs in terms of the domain insights.
iii. Phenotypes based on risk that act in prevention of the specified action plans.

Combined, the contributions presented cover all the research aims of effective detection, secure operation, transparent auditing, and actionable risk management.

6.2 Future Work

There are four paths that can take this work beyond high-fidelity simulation to production-level deployment. To begin with, simulated data should be substituted with streaming data between facilities, seasons, and suppliers in real-life setting validation. This allows periodically re-estimating the thresholds, updating site-specific clusters (phenotypes) and tracking domain drift.

Second, in the presence of time-stamped sensor sequences, temporal modelling with deep learning ought to be reconsidered. The sequence-aware architectures (e.g., LSTM/GRU, temporal convolution, or attention-based encoders) and temporal gradient boosting would be able to capture cumulative exposure, transient excursions, and recovery dynamics that would be inevitably lost on snapshot. This should be complemented by more rigorous probability calibration (e.g. isotonic) and uncertainty quantification (e.g. conformal prediction) in order to set cost-sensitive rules based on probability and intervals.

Third, formalisation is worthwhile to some cost-aware decision policies. Encoding asymmetric costs of missed contamination versus unnecessary recalls, laboratory capacity limits and delays, one can infer utility-maximising thresholds or policies customised to each site. Sample efficiency can be optimised by the active-learning loops being able to prioritise more labels when there is uncertainty or risk.

Fourth, feature space and data assurance are to increase: add microbiological assays, airflow/CO*2 measurements, packaging telemetry, more detailed handling records; normalise sensor QA (schedule, drift detection) and computerized data-quality checks (range, missingness, outlier guards). And lastly, apply explainability to counterfactuals (AKA minimal change to reduce risk) and causal analyses to prioritize interventions by how beneficial they are expected to be in terms of prevention.

These directions will maintain the strengths inherent to the technology accuracy, transparency, and phenotype-guided triage and enhance the robustness and value in an operational setting.

Overall, this thesis demonstrates that AI/ML, specifically XGBoost with SHAP explainability and clustering, can offer valid, explainable and actionable predictions of food contamination risk. The framework of the proposed framework makes it possible to achieve both scientific rigor and practical usability as it balances accuracy against operational safety (zero false positives) and interpretability. Though subsequent work is required to support the methods using actual, time-recorded data, the methods outlined here provide a scalable structure in which to base future, data-informed food safety management.

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